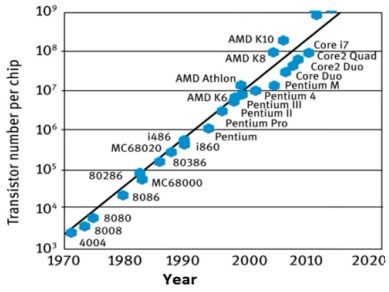
DIGITAL VLSI

(Very Large Scale Integration)

UE23EC252B

- Hitesh Pranav

Unit-1 Introduction to MOS Inverters



Moore's Law

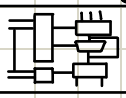
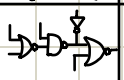
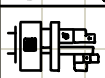
Moore's Law

→ The number of transistors per chip would grow exponentially (doubles every 18 months)

Synthesis Process

→ Converting HDL Code to optimised gate level design using Standard library

Levels of Design Abstraction

Design Level	System Design	Logic Design	Circuit Design	Layout Design
Graphical Representation				
Elements	Blocks, Sub-blocks	Logic Gates	Logic Gates	Geometric Structure

Design-Style Trade-offs (PPAs)

- Power Consumption
- Performance
- Design Time
- Unit Cost
- NRE Cost (Non-Recurring Engineering) (R&D)

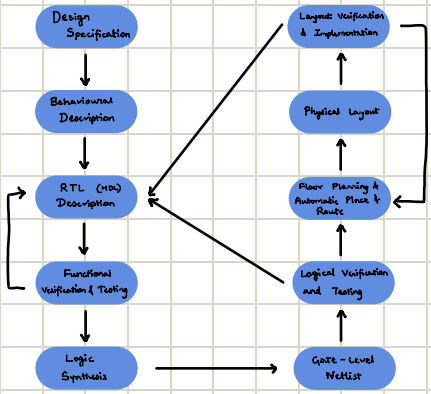
EDA Tools

- Cadence (Encounter RTL Compiler, Genus, Virtuoso)
- Mentor Graphics (Questa sim, Leonardo Spectrum, Tanner)
- Synopsys (Design Compiler, Gensys)
- Xilinx (ISim, Vivado, Xilinx Synthesis)

IRDS: International Roadmap for device & system

ASIC: Application Specific IC's

VLSI ASIC Design Flow



Challenges

- Big Data Analytics
- Feature Recognition
- IOT Applications
- Logic Device Scaling
- 3D NAND Flash Memory
- Cyber Physical Systems

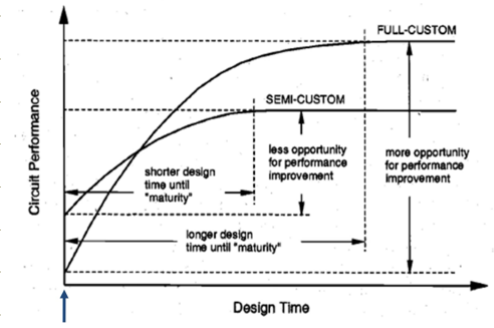
Overview of Design Methodology

→ Design complexity of logic chip increases almost exponentially with number of transistors to be integrated, this is translated into an increase into design cycle time

→ Design Cycle Time is the time period from start of chip development until the mask type delivery Most time is used to achieve desired level of performance at an acceptable cost

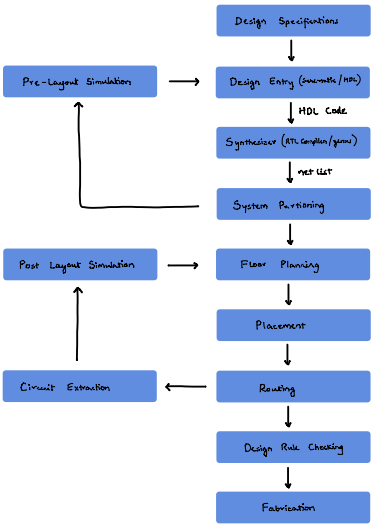
→ Design cycle time & achievable circuit performance depends on VLSI design style There are 2 types :

1) Full Custom Design Style
2) Semi Custom Design Style



Effect of design style on design cycle time & performance

Design Methodology



Full Custom

→ It requires longer time until the design maturity is reached

→ The flexibility in geometry & placement allows more opportunity for performance improvement

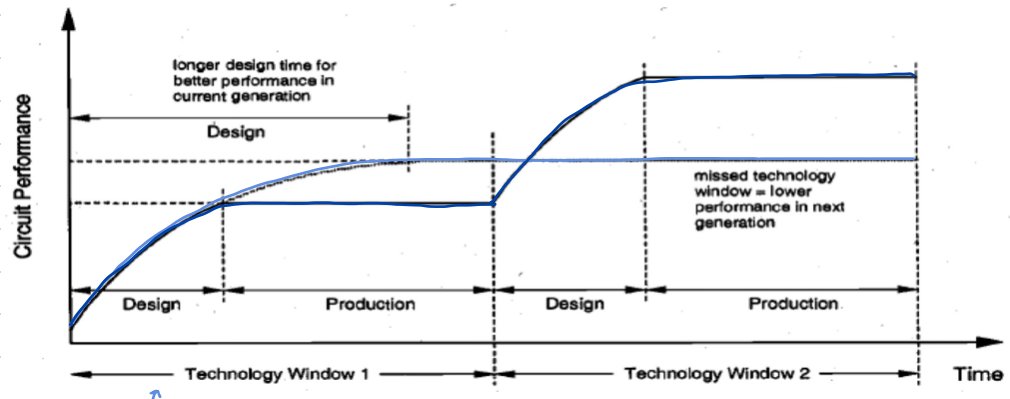
→ Final product has higher level of performance

→ Silicon area is relatively small because of better area utilization

Semi Custom

→ Less design time & less opportunity for performance improvement

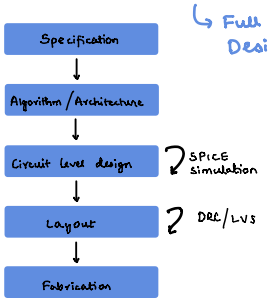
→ Design performance is Higher than Full custom



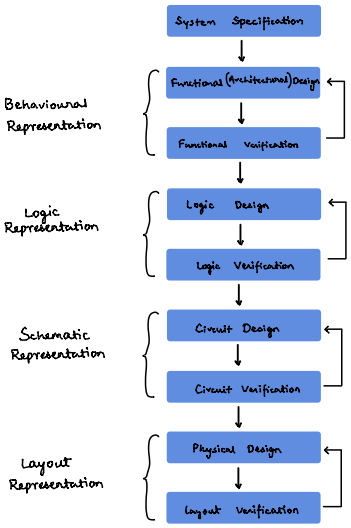
Effect of technology window on design time & performance

MOS Inverters (Static Characteristics)

↳ Full Custom Design



ASIC Design Flow ★



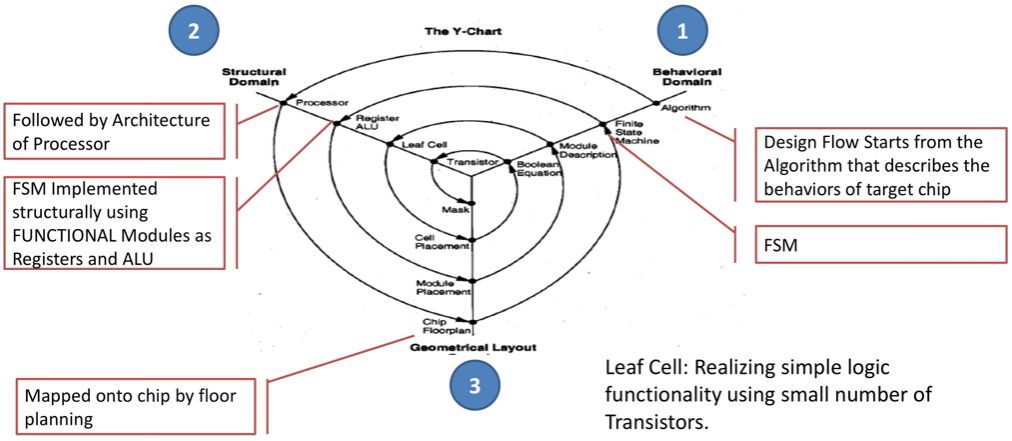
Full Custom	Semi Custom
1) Complete design, implementation placement is done from transition level	1) We use pre-design & pre-tested module for designing
2) High Cost	2) Low Cost
3) More design time required	3) Less design time required
4) High circuit performance	4) Low circuit performance
5) Less dependancy on exhisting technology	5) More dependancy on exhisting technology
6) Can be used for mass production	6) Can't be used for mass production
7) Very high speed	7) Less speed compared to Full Custom
8) Complex Circuit Layout	8) Simplified Circuit Layout
ex: High performance processes, FPGAs & memory chips	ex: STD Cell based, gate arrays etc.,

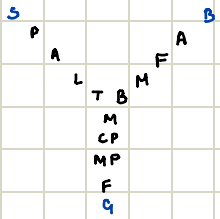
Y-Chart / Gajski Chart

→ It is a 3-part graphic organizer that is used for describing 3 aspects of a topic

Behavioural Domain → Structural Domain

Geometrical Layout





NOTE :

NMOS is a good transfer of logic 0

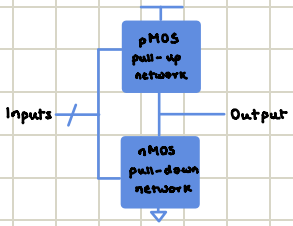
PMOS is a good transfer of logic 1

pull-up (PUN) ⇒ pulling to logic 1

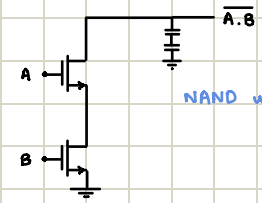
pull-down (PDN) ⇒ pulling to logic 0

General logic gate using pull-up and pull-down networks

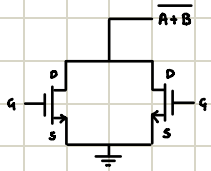
ex : CMOS Circuit



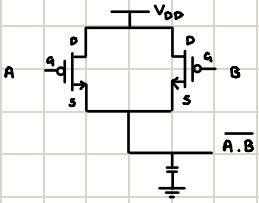
NAND and NOR Gates using Pull Down Network and Pull up Network



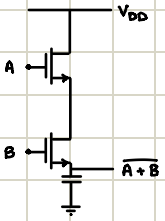
NAND using NMOS



NOR using NMOS



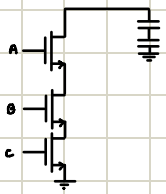
NAND using PMOS

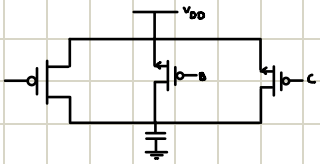


NOR using PMOS

Q. Find CMOS for $Y = \overline{A.B.C}$

A.



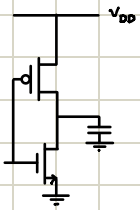


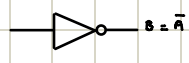
V_{Th} - Inverter Threshold Voltage

$V_{T,n}$ - NMOS V_{Th}

$V_{T,p}$ - PMOS V_{Th}

Inverter

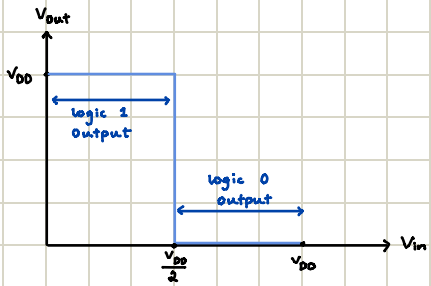




Symbol

A	B
0	1
1	0

Truth Table

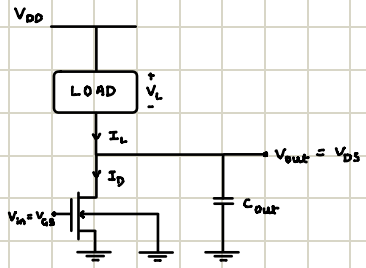


VTC of ideal inverter

NMOS Inverter

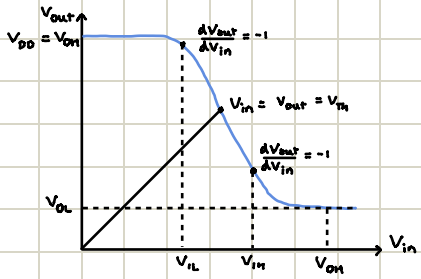
- → +ve load inverter
- → nMOSFET load inverter
  - → Depletion load inverter
  - → Enhancement load inverter
- → CMOS inverter

MOS Inverter Gate

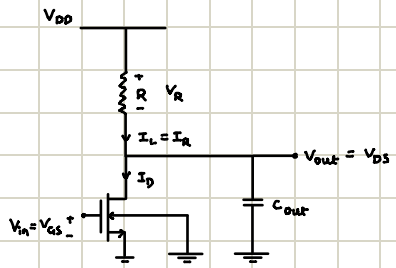


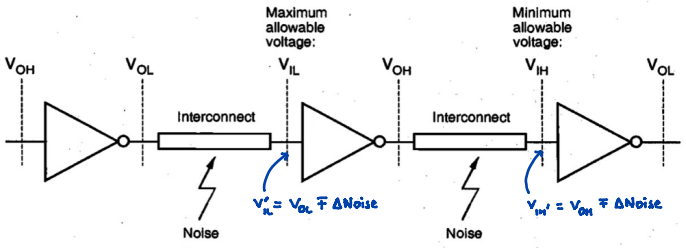
The load for inverter is considered as a circuit element in which 1 terminal is connected to V_{DD} & other to output. NMOS transistor is used as pull-down network in which drain is connected to V_{out} & source to ground. This is also called as driver transistor. Current entering the node V_{out} (I_L) = current leaving the node (I_D)

→ Voltage Transfer Characteristics

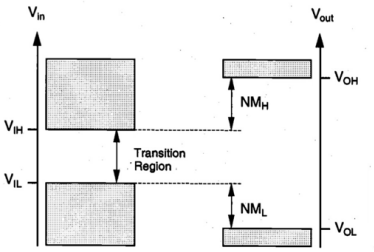


- V_{OH}: Max output voltage when output level is '1'
- V_{OL}: Min output voltage when output level is '0'
- V_{IL}: Max input voltage which can be interpreted as '0'
- V_{IH}: Min input voltage which can be interpreted as '1'





Noise Margin of MOS Inverter



Definition of Noise Margins

$N_{ML} = V_{IL} - V_{OL}$ (Noise Margin Low)

$N_{MH} = V_{OH} - V_{IH}$ (Noise Margin High)

- → In general ⇒ $V_{out} = f(V_{in})$
- → With Noise ⇒ $V'_{out} = f(V_{in} + \Delta V_{noise})$
- → Using Taylor Expansion ⇒ $V'_{out} = f(V_{in}) + \frac{dV_{out}}{dV_{in}} \Delta V_{noise}$ + higher order terms → neglected

Input for inverter 1

i) $< V_{IL}$ or b/w V_{OL} & V_{IL} ⇒ '0'

ii) $> V_{IL}$ ⇒ '1'

Input for inverter 2

i) $< V_{IH}$ ⇒ '1'

ii) $> V_{IH}$ or b/w V_{IH} & V_{OH}

→ Ideally, the slope of VTC should be very large between V_{IL} & V_{IH} because a narrow uncertain (transition) region allows for large noise margin. Reducing the width of the uncertain region is one of the important design objective

Perturbated o/p = Nominal o/p + (gain × External Perturbation)

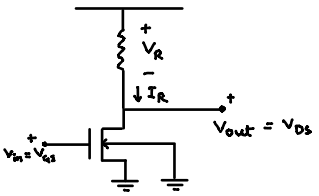
Types of Inverters

i) Resistive Load

ii) nMOS Load → Enhancement → Saturation / Linear; → Depletion

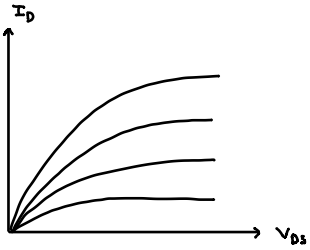
iii) CMOS Load

iv) BiCMOS Load



Resistive Load Inverter

- → Enhancement type nMOS is driver device
- → Load is simple linear resistance (R_L)
- → Power supply V_{DD}
- → I_D of driver MOS = I_L in DC s state
- → $V_{SB} = 0$ (No CLM)
- → V_{Th} is always equal to V_{T_0}



Cutoff ⇒ $V_{GS} < V_{Th}$, $I_D = 0$

Linear ⇒ $V_{GS} > V_{Th}$, $V_{DS} < V_{GS} - V_{Th}$, $I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{Th})V_{DS} - \frac{V_{DS}^2}{2}\right]$

$= \frac{K_n}{2}\left[2(V_{GS} - V_{Th})V_{DS} - V_{DS}^2\right]$

$= \frac{K_n}{2}\left[2(V_{in} - V_{Th})V_{out} - V_{out}^2\right]$

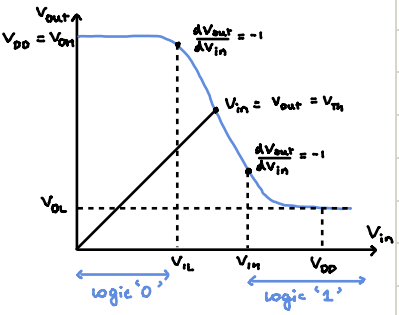
$K' = \mu_n C_{ox}$

$K_n = K' \frac{W}{L}$

Saturation ⇒ $V_{GS} > V_{Th}$, $V_{DS} \geq V_{GS} - V_{Th}$, $I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{Th})^2$

$= \frac{K_n}{2}(V_{in} - V_{Th})^2$

Input Voltage Range	Operating Mode
$V_{in} < V_{T_0}$	Cutoff
$V_{T_0} \leq V_{in} < V_{out} + V_{T_0}$	Saturation
$V_{in} \geq V_{out} + V_{T_0}$	Linear



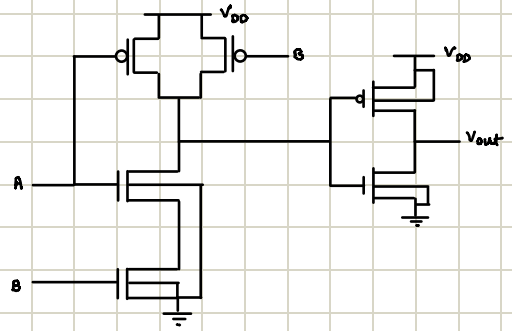
Note :

All Substrate to ground in NMOS

All Substrate to V_{DD} in PMOS

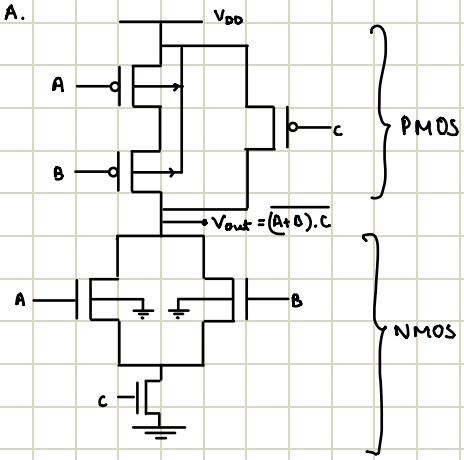
Q. 2 input AND Gate using CMOS Technology

A.



Q. $Y = \overline{(A+B).C}$

A.



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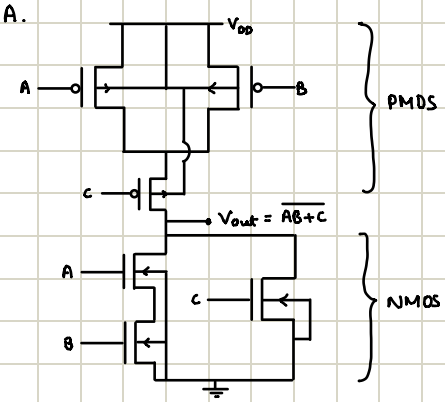
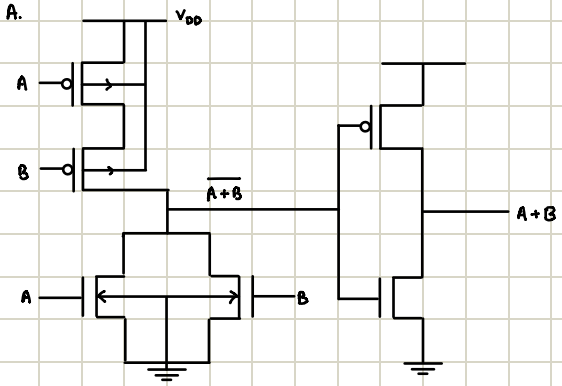
A.



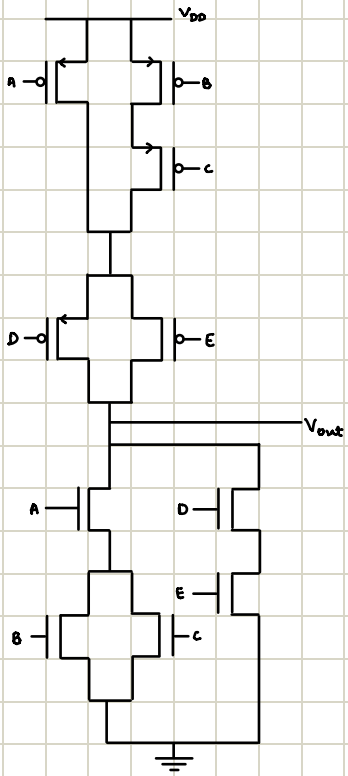
Q $\overline{\overline{A+B}}$

A.



Q. $\overline{A(B+C)+DE}$

A.



Power & Area Consideration

Power

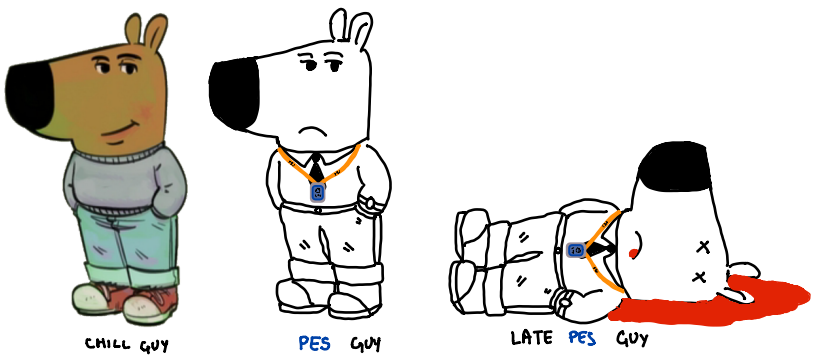
→ $P_{dc} = V_{DD} \times I_{DC}$

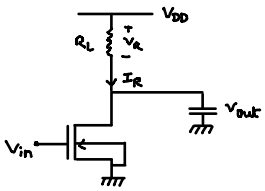
→ The DC power dissipation of inverter circuit in the product of power supply voltage & amount of current drawn from supply

→ $P_{DC} = \frac{V_{DD}}{2}\left[I_{DC}(V_{in=low}) + I_{DC}(V_{in=high})\right]$

Area

→ $\frac{W}{L}$ should be near unity in order to optimise the area but as it contradicts the other design criteria like noise margin, dynamic switching speed & driving capability





Calculating Critical Voltages

1) V_{OH} :

$V_{DD} - V_R - V_{out} = 0$

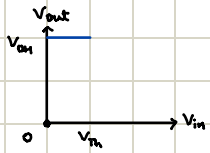
$V_{out} = V_{DD} - I_R R_L$

For $V_{in} < V_{Th}$, NMOS off $\Rightarrow I_D = 0$

WKT, $I_D = I_R = 0$

$V_{out} = V_{DD} = V_{OH}$

$V_{OH} = V_{DD}$ irrespective of resistive load value



2) V_{OL}

Consider $V_{in} = V_{OH} = V_{DD}$, device is in linear region of operation

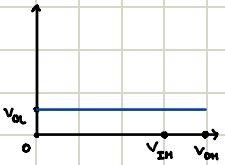
$V_{out} = V_{OL} = V_{DS}$

$I_D = I_R$

$I_R = \frac{V_{DD} - V_{out}}{R_L} = \frac{V_{DD} - V_{OL}}{R_L}$

$V_{GS} = V_{DD} > V_{Th}$ (ON)

$V_{DS} = V_{OL} < V_{DD} - V_{Th}$ (linear)



$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(2(V_{GS}-V_{Th})V_{DS} - V_{DS}^2\right)$$

$$= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left[2(V_{DD}-V_{Th})V_{OL} - V_{OL}^2\right]$$

$$= \frac{K_n}{2}\left[2(V_{DD}-V_{Th})V_{OL} - V_{OL}^2\right]$$

$$\Rightarrow \frac{V_{DD}-V_{OL}}{R_L} = K_n(V_{DD}-V_{Th})V_{OL} - \frac{K_n V_{OL}^2}{2}$$

$$\frac{K_n V_{OL}^2}{2} - K_n(V_{DD}-V_{Th}) - \frac{V_{OL}}{R_L} + \frac{V_{DD}}{R_L} = 0$$

$$\frac{K_n V_{OL}^2}{2} - \left(K_n(V_{DD}-V_{Th}) + \frac{1}{R_L}\right)V_{OL} + \frac{V_{DD}}{R_L} = 0$$

$$V_{OL}^2 - 2\left(V_{DD} - V_{Th} + \frac{1}{K_n R_L}\right)V_{OL} + \frac{2V_{DD}}{K_n R_L} = 0$$

$$V_{OL} = \frac{2\left(V_{DD} - V_{Th} + \frac{1}{K_n R_L}\right) - \sqrt{4\left(V_{DD} - V_{Th} + \frac{1}{K_n R_L}\right)^2 - \frac{8V_{DD}}{K_n R_L}}}{2}$$

$$= V_{DD} - V_{Th} + \frac{1}{K_n R_L} - \sqrt{\left(V_{DD} - V_{Th} + \frac{1}{K_n R_L}\right)^2 - \frac{2V_{DD}}{K_n R_L}}$$

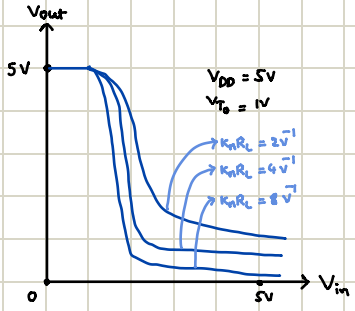
$V_{DD} \propto K_n R_L$

→ The value of output low voltage must be between 0 & V_{DD}

Out of the 2 possible solutions, the one which is physically correct for output low is choosen

→ The product '$\frac{1}{K_n R_L}$' is one of the important design parameter that determines value of V_{OL}

→ VTC for different value of parameter ($K_n R_L$)



V_{OL} for $2V^{-1} \Rightarrow 0.5V$

V_{OL} for $4V^{-1} \Rightarrow 0.3V$

V_{OL} for $8V^{-1} \Rightarrow 0.15V$

3) V_{IL}

$V_{in} = V_{IL}$ because $V_{DS} > V_{GS} - V_{Th} \Rightarrow$ NMOS in saturation

V_{IL} is smaller of the 2 input voltages at which slope VTC becomes -1 $\left(\frac{dV_{out}}{dV_{in}} = -1\right)$

$I_R = \frac{V_{DD} - V_{out}}{R_L}$

$I_D = \frac{K_n}{2}(V_{in} - V_{To})^2$

$\frac{V_{DD} - V_{out}}{R_L} = \frac{K_n}{2}(V_{in} - V_{To})^2$

Differentiate wrt $V_{in} \Rightarrow \frac{0 - \frac{dV_{out}}{dV_{in}}}{R_L} = K_n(V_{in} - V_{To})$

When $\frac{dV_{out}}{dV_{in}} = -1$, $V_{in} = V_{IL}$

$\frac{1}{R_L} = K_n(V_{IL} - V_{To})$

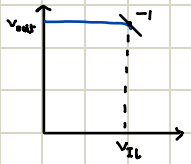
$V_{IL} = V_{To} + \frac{1}{K_n R_L}$

→ To determine output voltage when $V_{in} = V_{IL}$,

$I_R = I_D \Rightarrow \frac{V_{DD} - V_{out}}{R_L} = \frac{K_n}{2}(V_{in} - V_{To})^2$

$\frac{V_{DD} - V_{out}}{R_L} = \frac{K_n}{2}\left(V_{To} + \frac{1}{K_n R_L} - V_{To}\right)^2 = \frac{1}{2K_n R_L^2}$

$V_{DD} - V_{out} = \frac{1}{2K_n R_L} \Rightarrow V_{out} = V_{DD} - \frac{1}{2K_n R_L}$



Q. For $K_nR_L = 2/4/8$, $V_{DD} = 5V$, $V_{T_0} = 1V$, Find V_{IL}

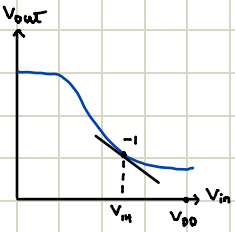
A. $V_{IL} = V_{T_0} + \frac{1}{K_nR_L}$

$(V_{IL})_{2V} = 1 + \frac{1}{2} = 1.5V \Rightarrow V_{out} = 5 - \frac{1}{4} = 4.75V$

$(V_{IL})_{2V} = 1 + \frac{1}{4} = 1.25V \Rightarrow V_{out} = 5 - \frac{1}{8} = 4.875V$

$(V_{T_L})_{8V} = 1 + \frac{1}{8} = 1.125V \Rightarrow V_{out} = 5 - \frac{1}{16} = 4.9375V$

4) V_{IH}



→ When $\frac{dV_{out}}{dV_{in}} = -1$, $V_{in} = V_{IH}$

→ NMOS in linear region

→ From KCL,

$I_{R_L} = I_D$ (linear)

$\frac{V_{DD} - V_{out}}{R_L} = \frac{K_n}{2}\left[2(V_{in} - V_{T_0})V_{out} - V_{out}^2\right]$

Differentiate wrt V_{in}

$\frac{-1}{R_L} \cdot \frac{dV_{out}}{dV_{in}} = \frac{K_n}{2}\left[2(V_{in} - V_{T_0})\frac{dV_{out}}{dV_{in}} + 2V_{out} - 2V_{out} \cdot \frac{dV_{out}}{dV_{in}}\right]$ ($V_{in} \to V_{IH}$)

Substitute $\frac{dV_{out}}{dV_{in}} = -1$ & $V_{in} = V_{IH}$

$\frac{1}{K_nR_L} = -V_{IH} + V_{T_0} + 2V_{out}$

→ To find V_{out} when $V_{in} = V_{IH}$

$\frac{V_{dd} - V_{out}}{R_L} = \frac{K_n}{2}\left[2\left(V_{T_0} + 2V_{out} - \frac{1}{K_nR_L} - V_{T_0}\right)V_{out} - V_{out}^2\right] = \frac{K_n}{2}\left[2\left(2V_{out} - \frac{1}{K_nR_L}\right)V_{out} - V_{out}^2\right]$

$V_{DD} - V_{out} = \frac{K_nR_L}{2}\left[2V_{out}^2 - \frac{2V_{out}}{K_nR_L} - \frac{V_{out}^2}{2}\right] = K_nR_L\left[\frac{3}{2}V_{out}^2 - \frac{V_{out}}{K_nR_L}\right]$

$V_{out}^2\left(\frac{3}{2}K_nR_L\right) - V_{out} - V_{DD} = 0 \Rightarrow V_{out} = \sqrt{\frac{2}{3} \cdot \frac{V_{DD}}{K_nR_L}}$ ($V_{out} \to 0$)

$V_{IH} = V_{T_H} + \sqrt{\frac{2V_{DD}}{3K_nR_L}} - \frac{1}{K_nR_L}$

Observation:

→ $V_{IH} \propto \frac{1}{K_nR_L}$

→ If K_nR_L increases, V_{OL} decreases

Note:

1) Higher the value of K_nR_L, smaller is the V_{IH} (logic 1 : larger range)

2) Higher the value of K_nR_L, smaller the V_{OL}.
(For ideal value of V_{OL} of inverter, K_nR_L should be longer)

3) V_{IL}, V_{IH}, V_{OH} are inversely proportional to K_nR_L

Summary

$V_{OH} = V_{DD}$

$V_{OL} = \left(V_{DD} - V_{T_0} + \frac{1}{K_nR_L}\right) - \sqrt{\left(V_{DD} - V_{T_0} + \frac{1}{K_nR_L}\right)^2 - \frac{2V_{DD}}{K_nR_L}}$

$V_{IL} = V_{T_0} + \frac{1}{K_nR_L}$

$V_{IH} = V_{T_0} + \sqrt{\frac{8V_{DD}}{3K_nR_L}} - \frac{1}{K_nR_L}$

Q. Consider the following inverter design problem : $V_{DD} = 5V$, $K_n = 30\mu A/V^2$ & $V_{T_0} = 1V$, design a resistive load inverter circuit with $V_{OL} = 0.2V$. Specifically, determine W/L ratio of the driver transistor and the value of load resistor that achieve required V_{OL}

A. $V_{OL} = 0.2V = V_{out}$

$V_{OH} = V_{DD} = 5V = V_{in} = V_{GS}$

$K' = \mu_n C_{ox}$

$K_n = \frac{K'W}{L}$

$V_{DS} = 0.2V < (V_{GS} - V_{TH})$ $\Rightarrow$ Linear region

$I_R = I_{D(linear)}$

$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{K_n'}{2}\frac{W}{L}\left[2(V_{OH} - V_{T_0})V_{OL} - V_{OL}^2\right]$$

$$\frac{4.8}{R_L} = \frac{30\times10^{-6}}{2}\cdot\frac{W}{L}\left[2(4)(0.2) - (0.2)^2\right]$$

$$\frac{L}{WR_L} = 4.875\times10^{-6}$$

$$\frac{WR_L}{L} = 2.051\times10^{5}\,\Omega = 205.1\,K\Omega$$

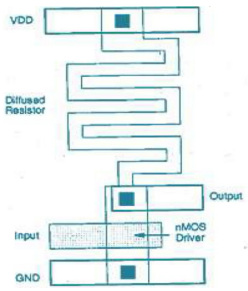
W/L Ratio	R_L (KΩ)	Power (mW) $P_{DC,avg}$
1	205	0.0585
2	102.5	0.117
3	68.33	0.175
4	51.25	0.234
5	41	0.2926
6	34.16	0.3512

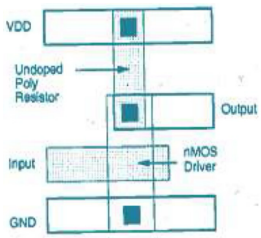
$$P_{DC} = \frac{V_{DD}}{2}\left[\cancel{I_{DC}\,(V_{in} = low)}^{\,0} + I_D\,(V_{in} = high)\right]$$

$$= \frac{V_{DD}}{2}\left[\frac{V_{DD} - V_{OL}}{R_L}\right]$$

→ Power consumption increases as R_L decreases & $\frac{W}{L}$ increases

→ If power consumption is the concern, $\frac{W}{L}$ is kept low & R_L is large





Chip Area

→ Chip area occupied by resistive load inverter depends on (W/L) ratio of NMOS driver transistor & R_L value

→ 2 Methods of resistor fabrication :

Diffused Resistor

→ Isolated n-type / p-type diffusion region with one contact on each end

→ The resistance is determined by density of diffusion region & the dimension $\left(\frac{L}{W}\text{ ratio}\right)$

→ Diffusion range, sheet resistance ⇒ 20 - 100 Ω/square

→ Large $\frac{L}{W}$ required to get resistor of (10-100)KΩ

Undoped Polysilicon Resistor

→ Polysilicon layers are heavily doped in order to reduce resistivity

The sheet resistivity of doped polysilicon interconnect & gate is 20 - 40 Ω/square

→ Drawback : R value of this can't be controlled very accurately which results in large variation in VTC

Q. Consider a resistive load inverter circuit with $V_{DD} = 5V$, $K_n' = 20\mu A/V^2$, $V_{To} = 0.8V$ $R_L = 200K\Omega$, $\frac{W}{L} = 2$. Calculate V_{OL}, V_{OH}, V_{IL}, V_{IH} on the VTC & find noise margins

A. $K_n R_L = K_n' R_L \times \frac{W}{L} = 4 \times 2 = 8$

$V_{OH} = V_{DD}$

$= 5V$

$$V_{OL} = \left(V_{DD} - V_{To} + \frac{1}{K_n R_L}\right) - \sqrt{\left(V_{DD} - V_{To} + \frac{1}{K_n R_L}\right)^2 - \frac{2V_{DD}}{K_n R_L}}$$

$$= \left(5 - 0.8 + \frac{1}{8}\right) - \sqrt{\left(5 - 0.8 + \frac{1}{8}\right)^2 - \frac{2\times 5}{8}}$$

$= 4.325 - 4.18$

$= 0.147\ V$

$V_{IL} = V_{To} + \frac{1}{K_n R_L}$

$= 0.8 + 0.125$

$= 0.925V$

$$V_{IH} = V_{To} + \sqrt{\frac{8V_{DD}}{3K_n R_L}} - \frac{1}{K_n R_L}$$

$$= 0.8 + \sqrt{\frac{8\times 5}{3\times 8}} - \frac{1}{8} = 1.966\ V$$

Noise Margin High ⇒ $NM_h = V_{DD} - V_{IH} = 5 - 1.966 = 3.034\ V$

Noise Margin Low ⇒ $NM_L = V_{IL} - V_{OL} = 0.925 - 0.147 = 0.778\ V$

For better noise immunity, noise margin for low signals should be 25% of V_{DD}

Q. Design resistive load inverter with $R = 1K\Omega$ such that $V_{OL} = 0.6V$.

The enhancement-type NMOS driver transistor has the following parameters:

$V_{DD} = 5V$, $V_{T_0} = 1V$, $\mu_n C_{ox} = 22\mu A/V^2$

a) Determine required W/L ratio

b) Determine V_{IL} & V_{IH}

c) Determine NM_L & NM_H

A. a) $$V_{OL} = \left(V_{DD} - V_{T_0} + \frac{1}{K_n R_L}\right) - \sqrt{\left(V_{DD} - V_{T_0} + \frac{1}{K_n R_L}\right)^2 - \frac{2V_{DD}}{K_n R_L}}$$

$$0.6 = \left(5 - 1 + \frac{1}{\frac{W}{L} \times 22 \times 10^{-6} \times 10^3}\right) - \sqrt{\left(5 - 1 + \frac{1}{\frac{W}{L} \times 22 \times 10^{-6} \times 10^3}\right)^2 - \frac{2 \times 5}{\frac{W}{L} \times 22 \times 10^{-6} \times 10^3}}$$

$$= \left(4 + \frac{500}{11\frac{W}{L}}\right) - \sqrt{\left(4 + \frac{500}{11\frac{W}{L}}\right)^2 - \left(\frac{5000}{11\frac{W}{L}}\right)}$$

$$\left(3.4 + \frac{500}{11\frac{W}{L}}\right)^2 = \left(4 + \frac{500}{11\frac{W}{L}}\right)^2 - \left(\frac{5000}{11\frac{W}{L}}\right)$$

$$\frac{5000}{11\frac{W}{L}} = \left(4 + \frac{500}{11\frac{W}{L}} + 3.4 + \frac{500}{11\frac{W}{L}}\right)\left(4 + \frac{500}{11\frac{W}{L}} - 3.4 - \frac{500}{11\frac{W}{L}}\right) \qquad (a^2 - b^2 = (a+b)(a-b))$$

$$\frac{5000}{11 \times 0.6\frac{W}{L}} = 7.4 + \frac{1000}{11\frac{W}{L}}$$

$$\frac{1000}{11\frac{W}{L}}\left(\frac{5}{0.6} - 1\right) = 7.4 \quad \Rightarrow \quad \frac{W}{L} = \frac{1000}{11 \times 7.4}\left(\frac{5}{0.6} - 1\right) = 90.1$$

b) $$V_{IL} = V_{T_0} + \frac{1}{K_n R_L} = 1 + \frac{1}{90.1 \times 22 \times 10^{-6} \times 10^3} = 1.504\,V$$

$$V_{IH} = V_{T_0} + \sqrt{\frac{8V_{DD}}{3K_n R_L}} - \frac{1}{K_n R_L}$$

$$= 1 + \sqrt{\frac{8 \times 5}{3} \times 0.504} - 0.504 \simeq 2V$$

c) $$NM_L = V_{IL} - V_{OL}$$

$$= 1.504 - 0.6 = 0.904V$$

$$NM_H = V_{OH} - V_{IH}$$

$$= 5 - 2 = 3V$$

Area

→ W/L should be near unity in order to optimize the area but as it contradicts the other design criteria like noise margin, dynamic switching speed & driving capability (trade off)

NMOS Load Inverters

→ Large area occupied by load resistor invertor, hence not suitable for digital circuits

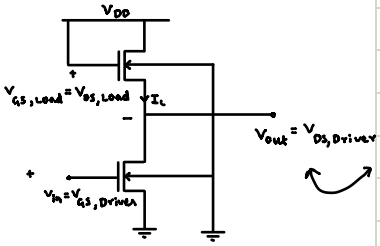
→ nMOS transistor as active load device

→ Main advantage of using MOSFET as load device is that the silicon area occupied by transistor is usually smaller than that occupied by resistive load

→ 2 Types :

i) Enhancement - Load NMOS Inverter

ii) Depletion - Load NMOS Inverter



Enhancement - Load nMOS Inverter (no channel)

→ Depending on bias voltage applied to its gate terminal, the load transistor can be operated either in saturation or linear region

→ Saturated enhancement type nMOS Load

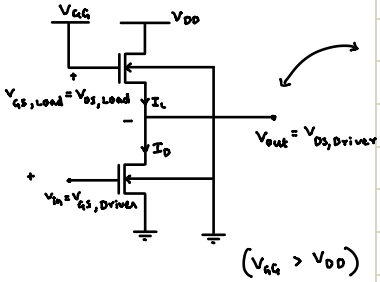
→ Required single voltage supply & relatively simple fabrication process, yet VOH level is limited to $V_{DD} - VT_{Load}$ ($V_{DS} = V_{GS}$)

→ Linear enhancement type nMOS Load

→ $V_{OH} = V_{DD}$ ⇒ higher noise margins compared to saturated load inverter ($V_{GS} >> V_{DS}$)

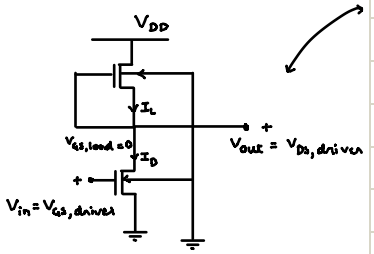
→ Drawback is that it uses 2 power supply voltages

→ Both suffer from relatively high stand-by (DC) power dissipation



($V_{GG} > V_{DD}$)

Depletion - Load nMOS Inverter (channel pre-fabricated)



→ Fabrication process is slightly complicated in depletion load inverter channel implant to adjust threshold voltage of load device

→ Advantages of Depletion load nMOS inverter

i) sharp VTC transition & better noise margins

ii) Single power supply

iii) Smaller overall layout area

→ $V_{T_{load}}$ = -ve , $V_{T_{driver}}$ = +ve

→ Single p-substrate (both driver & load)

→ Channel is already present

→ $V_{GS} = 0$, NMOS is already ON

→ V_{SB} of depletion load is V_{out}

→ $V_{T_{Load}}$ is a function of V_{out}

$V_{SB_{driver}} = 0$

→ $V_{T,load} = V_{TO,load} + \gamma\left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|}\right)$

→ Device in saturation : $I_D = \frac{K_n}{2}(V_{GS} - V_{Th})^2$

$$I_{D,load} = \frac{K_{n,load}}{2}\left[-V_{T,load}(V_{out})\right]^2$$

$$= \frac{K_{n,load}}{2}\left|V_{T,load}(V_{out})\right|^2$$

Device in linear : $I_D = \frac{K_n}{2}\left[2(V_{GS} - V_{Th})V_{DS} - V_{DS}^2\right]$

$$I_{D,load} = \frac{K_{n,load}}{2}\left[2\left|V_{T,load}(V_{out})\right|(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2\right]$$

For Depletion Load, $V_{GS} = 0V$

$V_{DS} = V_{DD} - V_{out}$

$V_{SB} = V_{out}$

$V_T = -Ve$

$V_{T,load}$ → Function of V_{out}

channel already formed

always ON ($V_{GS} = 0$)

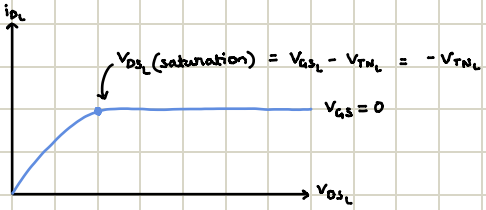
Enhancement driver, $V_{GS} = V_{in}$

$V_{DS} = V_{out}$

$V_{T,driver} = V_{T_0}$

$V_{SB} = 0V$

$V_T = +ve$



→ Average DC Power ⇒ $P_{DC} = V_{DD} \times I_{DC}$

$$= \frac{V_{DD}}{2}\left[\cancel{I_{DC}(V_{in}=0)}^{\,0} + I_{DC}(V_{in} = V_{OH})\right]$$

For input high, load → saturation (I_{DC_L})

driver → linear (I_{DC_N})

→ $P_{DC} = \frac{V_{DD}}{2}\left[I_{DL(sat)}\right]$ ($V_{GS} = 0$, $V_T = V_{T_{load}}(V_{OL})$)

$P_{DC} = \frac{V_{DD}}{2}\left[I_{DN}(lin)\right]$ ($V_{GS} = V_{in} = V_{OH}$, $V_T = V_{Th}$, $V_{DS} = V_{out} = V_{OL}$)

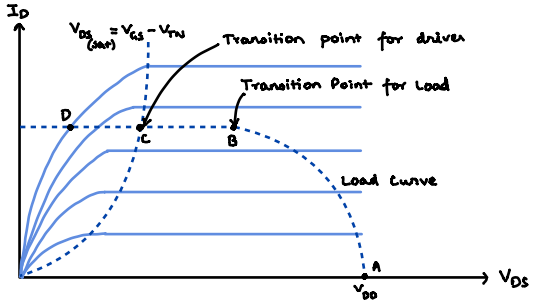
$$P_{DC} = \frac{V_{DD}}{2}\left[\frac{K_{n,load}}{2}\left|V_{T,load}(V_{out})\right|^2\right]$$

$$= \frac{1}{4} \cdot V_{DD} \cdot K_{n,load}\left|V_{T,load}(V_{out})\right|^2$$

$$\rightarrow P_{DC} = \frac{V_{DD}\,K_{nL}}{4}\left|V_{T,load}(V_{OL})\right|^2$$

$$P_{DC} = \frac{V_{DD}}{2}\left[\frac{K_n}{2}\left[2(V_{OH} - V_{Th}) - V_{OL}^2\right]\right] = \frac{V_{DD}K_n}{4}\left[2(V_{OH} - V_{Th})V_{OL} - V_{OL}^2\right]$$

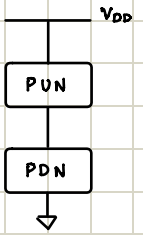
V_{in}	V_{out}	Driver	Load
V_{OL}	V_{OH}	cutoff	linear
V_{IL}	$\simeq V_{OH}$	saturation	linear
V_{OH}	V_{OL}	linear	saturation
V_{IH}	$\simeq$ small	linear	saturation



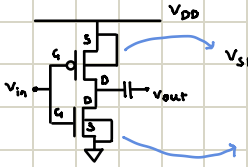
Merging the two transistor output characteristics when input V_{in} changes from V_{OL} to V_{OH}

CMOS Inverter

→



$\Rightarrow$



$V_{SB} = 0 \Rightarrow V_{T_0,P}$ → Load / Pull up Transistor

$V_{SB} = 0 \Rightarrow V_{T_0,n}$ → Driver / Pull Down Transistor

→ Logic 0 to PMOS $\Rightarrow$ ON

Logic 1 to NMOS $\Rightarrow$ ON

PMOS is good conductor of Logic 0

NMOS is good conductor of Logic 1

V_{in}	V_{out}
0	1
1	0

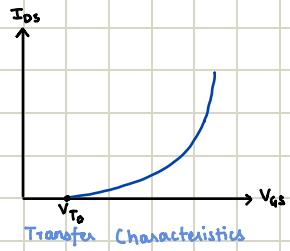
→ Case (i) ($V_{in} = 0$, PMOS ON, NMOS OFF)

$V_{in} = 0$, $V_{out} = V_{DD}$

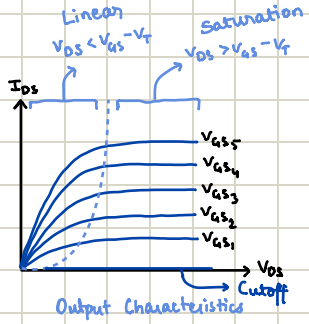
Case (ii) ($V_{in} = 1$, PMOS OFF, NMOS ON)

$V_{in} = V_{DD}$, $V_{out} = 0$

→ For NMOS,

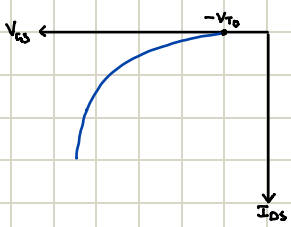


Transfer Characteristics

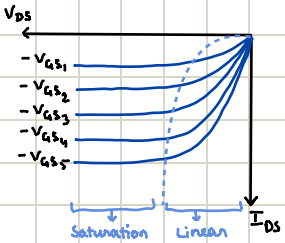


Output Characteristics

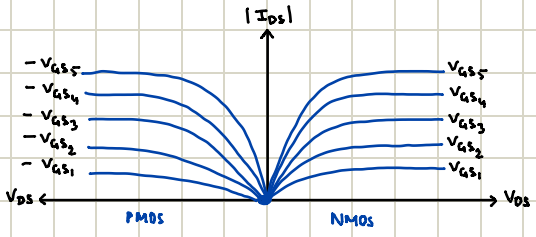
→ For PMOS,



Transfer Characteristics



Output Characteristics



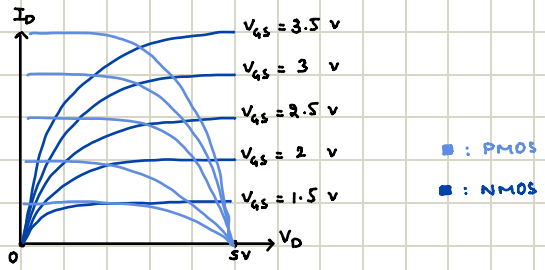
→ Take CMOS Inverter,

$V_{DD} = 5V$

$V_{T_0,n} = 1V$

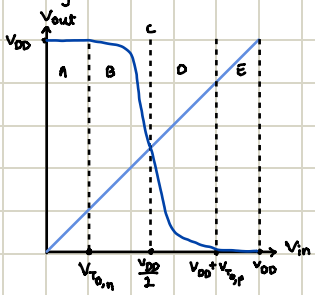
$V_{T_0,p} = -1V$

$V_{T_0,n} = |V_{T_0,p}| = 1V$



Superimposed NMOS & PMOS

Using this we draw VTC



Region	Input (V_{in})	Output (V_{out})	PMOS	NMOS
I	V_{OL}	V_{OH}	Linear	OFF
II	$\simeq V_{OL}$	$< V_{OH}$	Linear	Saturation
III	$V_{DD}/2$	$V_{DD}/2$	Saturation	Saturation
IV	$\simeq V_{OH}$	$> V_{OH}$	Saturation	Linear
V	V_{OH}	V_{OL}	OFF	Linear

Advantages

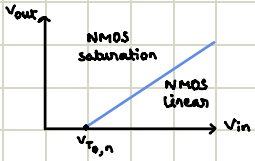
→ Steady state power dissipation of CMOS inverter is virtually negligible, except for small power dissipation due to leakage current

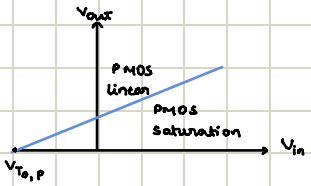
→ VTC exhibits a full output voltage swing between 0 & V_{DD} and that the VTC transition is usually very sharp

Disadvantages

→ 2 seperate substrate is required

→ More Area





Case (i) : $(V_{in} < T_{on})$

→ nMOS OFF

→ pMOS ON (linear region)

→ $I_{D,n} = I_{D,p} = 0$

The D to S voltage of pMOS is 0, $V_{out} = V_{OH} = V_{DD}$

Case (ii) : $(V_{in} > V_{DD} + V_{Top})$

→ pMOS OFF

→ nMOS ON linear region

→ $V_{DS_n} = 0V$

$V_{out} = V_{OL} = 0V$

Case (iii) : $(V_{in} = V_{IL})$

→ nMOS ON saturation

→ pMOS ON linear

→ Since $I_{Dn} = I_{Dp}$,

$$\frac{K_n}{2}(V_{in} - V_{Ton})^2 = \frac{K_p}{2}\left[2(V_{GSp} - V_{Top})V_{DSp} - V_{DSp}^2\right]$$

$$K_n(V_{in} - V_{Ton})^2 = K_p\left[2(V_{in} - V_{DD} - V_{Top})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2\right]$$

Differentiating wrt V_{in}

$$2K_n(V_{in} - V_{Ton}) = K_p\left[2(V_{in} - V_{DD} - V_{Top})\frac{\partial V_{out}}{\partial V_{in}} + 2(V_{out} - V_{DD}) - 2(V_{out} - V_{DD})\frac{\partial V_{out}}{\partial V_{in}}\right]$$

$\left(\frac{\partial V_{out}}{\partial V_{in}} = -1 \Rightarrow V_{in} = V_{IL}\right)$

$$K_n(V_{IL} - V_{Ton}) = K_p\left[-V_{IL} + V_{DD} + V_{Top} + V_{out} - V_{DD} + V_{out} - V_{DD}\right]$$

$$\frac{K_n}{K_p}(V_{IL} - V_{Ton}) + V_{IL} - V_{Top} + V_{DD} = 2V_{out} \Rightarrow \frac{K_p(2V_{out} + V_{Top} - V_{DD}) + K_n V_{Ton}}{K_n + K_p} = V_{IL}$$

$$V_{IL} = \frac{(2V_{out} + V_{Top} - V_{DD}) + \frac{K_n}{K_p}V_{Ton}}{1 + \frac{K_n}{K_p}}$$

$$V_{IL} = \frac{(2V_{out} + V_{Top} - V_{DD}) + K_r V_{Ton}}{1 + K_r}$$

Case (iv): $(V_{in} = V_{IH})$

→ nMOS ON linear

pMOS ON saturation

→ I_{D_n} (linear) = I_{D_p} (saturation)

$$\frac{K_n}{2}\left[2(V_{GS_n} - V_{Th})V_{DS} - V_{DS}^2\right] = \frac{K_p}{2}\left[V_{GS_p} - V_{T_p}\right]^2$$

$$\frac{K_n}{2}\left[2(V_{in} - V_{T_{on}})V_{out} - V_{out}^2\right] = \frac{K_p}{2}\left[(V_{in} - V_{DD}) - V_{T_{op}}\right]^2$$

$$\frac{K_n}{2}\left[(V_{in} - V_{T_{on}})\frac{\partial V_{out}}{\partial V_{in}} + V_{out} - V_{out}\frac{\partial V_{out}}{\partial V_{in}}\right] = \frac{K_p}{2} \times 2\left(V_{in} - V_{DD} - V_{T_{op}}\right)$$

$$K_n\left[-V_{IH} + V_{T_{on}} + V_{out} + V_{out}\right] = K_p\left(V_{IH} - V_{DD} - V_{T_{op}}\right) \qquad \left(\frac{\partial V_{out}}{\partial V_{in}} = -1 \Rightarrow V_{in} = V_{IH}\right)$$

$$\frac{K_n}{K_p}\left[2V_{out} - V_{IH} + V_{T_{on}}\right] = V_{IH} - V_{DD} - V_{T_{op}}$$

$$\frac{K_n}{K_p}\left[2V_{out} + V_{T_{on}}\right] + V_{DD} + V_{T_{op}} = \left(1 + \frac{K_n}{K_p}\right)V_{IH}$$

$$V_{IH} = \frac{K_r\left(2V_{out} + V_{T_{on}}\right) + V_{DD} + V_{T_{op}}}{1 + K_r}$$

Case (v): Region C, $V_{in} = V_{Th}$

→ Both nMOS & pMOS in saturation

I_{D_n} (saturation) = I_{D_p} (saturation)

$$\frac{K_n}{2}\left[V_{GS} - V_{T_{on}}\right]^2 = \frac{K_p}{2}\left[V_{GS_p} - V_{T_{op}}\right]^2$$

$$K_n\left(V_{in} - V_{T_{on}}\right)^2 = K_p\left(V_{in} - V_{DD} - V_{T_{op}}\right)^2$$

$$(V_{in} - V_{T_{on}}) = \pm\sqrt{\frac{K_p}{K_n}}\left(V_{in} - V_{DD} - V_{T_{op}}\right)$$

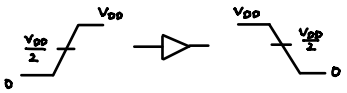
-ve is valid, +ve gives denomitor as 0

$$V_{in} - V_{T_{on}} = -\sqrt{\frac{1}{K_r}}V_{in} + \sqrt{\frac{1}{K_r}}\left(V_{DD} + V_{T_{op}}\right)$$

$$V_{in} = V_{Th} = \frac{V_{T_{on}} + \sqrt{\frac{1}{K_r}}\left(V_{DD} + V_{T_{op}}\right)}{1 + \sqrt{\frac{1}{K_r}}}$$

→ For ideal CMOS inverter, $V_{T_{on}} = 1V$, $V_{T_{op}} = -1V$, $\frac{K_p}{K_n} = 1$

$$V_{Th} = \frac{1 + (V_{DD} - 1)}{1 + 1} = \frac{V_{DD}}{2}$$



→ For symmetric CMOS inverter

$$K_r = \frac{K_n}{K_p} = 1 \Rightarrow \frac{\mu_n\left(\frac{W}{L}\right)_n}{\mu_p\left(\frac{W}{L}\right)_p} = 1$$

And typically, $1 = \frac{580\left(\frac{W}{L}\right)_n}{230\left(\frac{W}{L}\right)_p} \Rightarrow \left(\frac{W}{L}\right)_p \simeq 2.5\left(\frac{W}{L}\right)_n$

Summary,

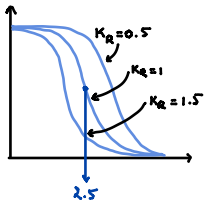
→ $V_{OL} = 0V \quad \left[I_{D_n} = I_{D_p} = 0\right]$

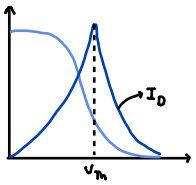
→ $V_{OH} = V_{DD}$

→ $V_{IL} = \frac{(2V_{out} + V_{T_{op}} - V_{DD}) + K_r V_{T_{on}}}{1 + K_r} \quad \left[I_{D_n}(\text{saturation}) = I_{D_p}(\text{linear})\right]$

→ $V_{IH} = \frac{K_r\left(2V_{out} + V_{T_{on}}\right) + V_{DD} + V_{T_{op}}}{1 + K_r} \quad \left[I_{D_n}(\text{linear}) = I_{D_p}(\text{saturation})\right]$

→ $V_{Th} = \frac{V_{T_{on}} + \sqrt{\frac{1}{K_r}}\left(V_{DD} + V_{T_{op}}\right)}{1 + \sqrt{\frac{1}{K_r}}} \quad \left[I_{D_n}(\text{saturation}) = I_{D_p}(\text{saturation})\right]$





Q. Assume $V_{DD} = 5V$, $V_{T_n} = 1V$, $V_{Tp} = -1V$

$K_R = 0.5, 1, 1.5$. Determine V_{Th}

A. at $K_R = 0.5$, $V_{Th} = \frac{1 + \sqrt{\frac{1}{0.5}}(5-1)}{1+\sqrt{\frac{1}{0.5}}} = 2.757$

at $K_R = 1$, $V_{Th} = \frac{1 + \sqrt{\frac{1}{1}}(5-1)}{1+\sqrt{\frac{1}{1}}} = 2.5$

at $K_R = .5$, $V_{Th} = \frac{1 + \sqrt{\frac{1}{1.5}}(5-1)}{1+\sqrt{\frac{1}{1.5}}} = 2.348$

→ As size of pMOS is increased greater than 2.5 times, the VTC curve shifts towards right

→ As size of pMOS is decreased lesser than 2.5 times, the VTC curve shifts towards left

→ I_D max at V_{Th}

→ Increase in $K_R > 1$, V_{Th} decreases less than $\frac{V_{DD}}{2}$

→ Decrease in $K_R < 1$, V_{Th} increases more than $\frac{V_{DD}}{2}$

Q. $V_{DD} = 3V$ $V_{T_{0n}} = 0.7V$ $V_{T_{0p}} = -0.7V$

$K_R = 1, 4, 0.25$. $V_{Th} = ?$

A. $K_R = 1$, $V_{Th} = \frac{0.7 + \sqrt{\frac{1}{1}}(3-0.7)}{1+\sqrt{\frac{1}{1}}} = 1.5$

$K_R = 4$, $V_{Th} = \frac{0.7 + \sqrt{\frac{1}{4}}(3-0.7)}{1+\sqrt{\frac{1}{4}}} = 1.23V$

$K_R = 0.25$, $V_{Th} = \frac{0.7 + \sqrt{\frac{1}{0.25}}(3-0.7)}{1+\sqrt{\frac{1}{0.25}}} = 1.77V$

Q. Consider a CMOS inverter circuit with following parameters:

$V_{DD} = 3.3V$, $V_{T_{0n}} = 0.6V$, $V_{T_{0p}} = -0.7V$, $K_n = 200\mu A/V^2$, $K_p = 80\mu A/V^2$. Calculate Noise Margins

A. $V_{OH} = V_{DD} = 3.3V$; $V_{OL} = 0V$

$$V_{IL} = \frac{(2V_{out} + V_{Top} - V_{DD}) + K_R V_{T0n}}{1+K_R} = \frac{(2V_{out} - 0.7 - 3.3) + 0.6\left(\frac{200}{80}\right)}{1+\frac{200}{80}} = \frac{2V_{out} - 2.5}{3.5}$$

at $V_{in} = V_{IL}$, $I_{D_n}(sat) = I_{D_p}(lin)$

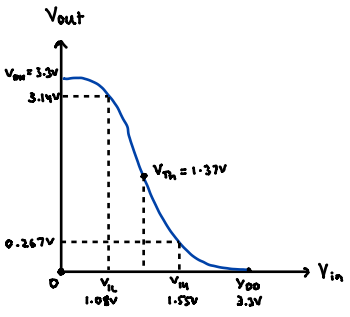
$$\frac{K_n}{2}(V_{in} - V_{T_{0n}})^2 = \frac{K_p}{2}\left[2(V_{in} - V_{DD} - V_{Top})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2\right]$$

$$2.5\left(\frac{2V_{out}}{3.5} - \frac{2.5}{3.5} - 0.6\right)^2 = \left[2\left(\frac{2V_{out}}{3.5} - \frac{2.5}{3.5} - 3.3 + 0.7\right)(V_{out} - 3.3) - (V_{out} - 3.3)^2\right] \Rightarrow 0.672 V_{out}^2 + 0.03 V_{out} - 6.66 = 0$$

By solving quadratic eqn, $V_{out} = 3.14$ or ~~-3.17~~

$$V_{IL} = \frac{2V_{out}}{3.5} - \frac{2.5}{3.5} = 1.08V$$

$$V_{IH} = \frac{K_R(2V_{out} + V_{T_{0n}}) + V_{DD} + V_{Top}}{1+K_R} = \frac{2.5(2V_{out} + 0.6) + 3.3 - 0.7}{1+2.5} = 1.42V_{out} + 1.17\ V$$



At V_{IH}, $I_{D_n}(lin) = I_{D_p}(sat)$

$$K_n\left[2(V_{in} - V_{T_{0n}})V_{out} - V_{out}^2\right] = K_p\left[V_{in} - V_{DD} - V_{Tp}\right]^2$$

$$2.5\left[2(1.42V_{out} + 1.17 - 0.6)V_{out} - V_{out}^2\right] = (1.42V_{out} + 1.17 - 3.3 + 0.7)^2$$

$$2.59 V_{out}^2 + 6.91 V_{out} - 2.044 = 0$$

$V_{out} = 0.268$ or ~~-3.36~~

$V_{in} = 1.42(0.268) + 1.17 = 1.55V$

$NM_L = V_{IL} - V_{OL} = 1.08 - 0 = 1.08V$

$NM_H = V_{OH} - V_{in} = 3.3 - 1.55 = 1.75V$

$K_R > 1 \Rightarrow$ Not symmetric, $\frac{V_{DD}}{2} > V_{TH}$, which is true $(1.65 > 1.37)$

Q. Consider CMOS inverter

nMOS → $V_{T_{on}} = 0.6V$ $\quad \mu_n C_{ox} = 60\mu A/V^2$ $\quad (W/L)_n = 8$

pMOS → $V_{T_{op}} = -0.7V$ $\quad \mu_p C_{ox} = 25\mu A/V^2$ $\quad (W/L)_p = 12$

Calculate Noise margin & Switching Threshold. $\quad V_{DD} = 3.3V$

A. $K_r = \frac{60 \times 8}{25 \times 12} = 1.6$

$V_{OL} = 0V$, $V_{OH} = 3.3V$

$$V_{IL} = \frac{(2V_{out} + V_{T_{op}} - V_{DD}) + K_r V_{T_{on}}}{1 + K_r} = \frac{(2V_{out} - 0.7 - 3.3) + (1.6 \times 0.6)}{1 + 1.6} = \frac{(2V_{out} - 3.04)}{2.6}$$

$V_{in} = V_{IL} \Rightarrow I_{D_n}(sat) = I_{D_p}(lin)$

$$\frac{K_n}{2}(V_{in} - V_{T_{on}})^2 = \frac{K_p}{2}\left[2(V_{in} - V_{DD} - V_{T_{op}})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2\right]$$

$$1.6\left(\frac{2V_{out}}{2.6} - \frac{3.04}{2.6}\right)^2 = \left[2\left(\frac{2V_{out}}{2.6} - \frac{3.04}{2.6} - 3.3 + 0.7\right)(V_{out} - 3.3) - (V_{out} - 3.3)^2\right]$$

$$0.54V_{out}^2 - 7.25V_{out} + 15.66 = 0$$

$V_{out} = 10.92$ or 2.7

$$V_{IL} = \frac{2(2.7) - 3.04}{2.6} = 0.907$$

$$V_{IH} = \frac{K_r(2V_{out} + V_{T_{on}}) + V_{DD} + V_{T_{op}}}{1 + K_r} = \frac{1.6(2V_{out} + 0.6) + 3.3 - 0.7}{1 + 1.6} = \frac{3.2V_{out} + 3.56}{2.6} = 1.23V_{out} + 1.37$$

At V_{IH}, $I_{D_n}(lin) = I_{D_p}(sat)$

$$K_n\left[2(V_{in} - V_{T_{on}})V_{out} - V_{out}^2\right] = K_p\left[V_{in} - V_{DD} - V_{T_{op}}\right]^2$$

$$1.6\left[2(1.23V_{out} - 1.37 - 0.6)V_{out} - V_{out}^2\right] = \left[1.23V_{out} - 1.37 - 3.3 + 0.7\right]^2$$

$$0.54V_{out}^2 - 6.16V_{out} - 1 = 0$$

$V_{out} = 0.16$ or -11.5

$V_{IH} = 1.23V_{out} + 1.37 = 1.57V$

$$V_{Th} = \frac{V_{T_{on}} + \sqrt{\frac{1}{K_r}}(V_{DD} + V_{T_{op}})}{1 + \sqrt{\frac{1}{K_r}}} = \frac{0.6 + \sqrt{\frac{1}{1.6}}(3.3 - 0.7)}{1 + \sqrt{\frac{1}{1.6}}} = 1.48V$$

$NM_L = V_{IL} - V_{OL} = 0.907V$

$NM_H = V_{OH} - V_{IH} = 1.73V$

Q. Design resistive-load inverter with $R = 1K\Omega$, $V_{OL} = 0.6V$, $V_{DD} = 5V$, $\gamma = 0.2V^{1/2}$, $\mu_n C_{ox} = 22\mu A/V^2$, $V_{T_0} = 1V$, $\lambda = 0$. Find a) W/L b) V_{IL} & V_{IH} c) NM_L & NM_H

A. a) $I_R = I_{D_n}(lin) \Rightarrow \frac{V_{DD} - V_{out}}{R_L} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left[2(V_{in} - V_{T_0})V_{out} - V_{out}^2\right] \Rightarrow 400 = \frac{W}{L}\left[2(5-1)(0.6) - (0.6)^2\right] \Rightarrow \frac{W}{L} = 90.1$

b) $V_{IL} = 1 + \frac{1}{1000 \times 1.98} = 1.5V$, $V_{IH} = 3.09V$ $\left(V_{IH} = V_{T_0} + \sqrt{\frac{8V_{DD}}{3K_nR_L}} - \frac{1}{K_nR_L}\right)$ $\left(V_{IL} = V_{T_0} + \frac{1}{K_nR_L}\right)$

c) $NM_L = 1.5 - 0.6 = 0.9V$; $NM_H = 5 - 3.09 = 1.92V$

$P_{DC} = V_{DD} I_{DC}$ $\quad \left(I_R = I_{DC} = \frac{5 - 0.6}{1000} = 4.4mA\right)$

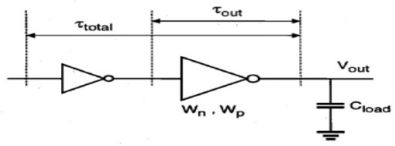
$= 5 \times 4.4m$

$= 22mW$

Bi CMOS Circuit

→ Signal propagation delay due to large interconnected capacitances is major factor that limits the performance of CMOS digital integrated circuits

→ System speed is restricted by current - driving capacity of CMOS gates

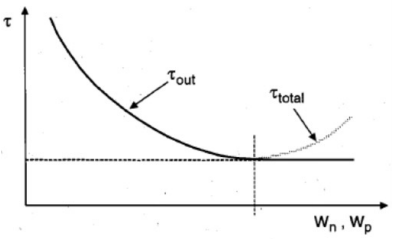


2nd inverter is larger to increase the I to increase performance

2 Stage CMOS inverter

→ Driving the large capacitance load

→ By increasing size of W of 2nd CMOS inverter improves driving capability but overall delay also increases for larger value of W_p & W_n because gate capacitance is added to 1st stage CMOS



→ Major Drawback of CMOS inverter
  → Low Driving capability for large capacitance loads
  → For driving large capacitive loads if scaled buffer chain is used, then larger amounts of Si required

→ Advantages of CMOS inverter
  → Low static power dissipation
  → Bidirectional Capability (Source & Drain interchangable)
  → Full swing output ($V_{OH} = V_{DD}$ & $V_{OL} = 0V$)

BJTs: Have large driving capability but have more power dissipation

MOS: Have low power dissipation

BiCMOS: use driving capability of BJT and low power dissipation of MOS to drive large capacitive load

BiCMOS Inverter (with resistive base pull-down)

→ For small base current, switch is closed. Large current flows

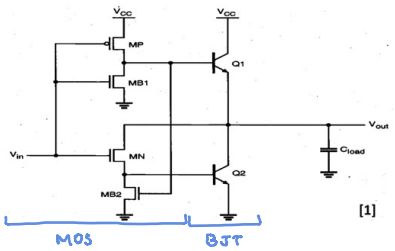
→ The complementary pMOS & nMOS transistor M_p & M_N supply base currents to BJT & thus act as trigger devices for bipolar output stage

→ Q_1 can efficiently pull up output voltage in presence of large output capacitance

→ Q_2 pulls down output voltage

→ Depending on V_{in}, either M_N or M_p can be ON in steady state, ensuring push pull operation for BJT

→ Resistance are used to remove the base charge of BJT when they are in cut-off mode

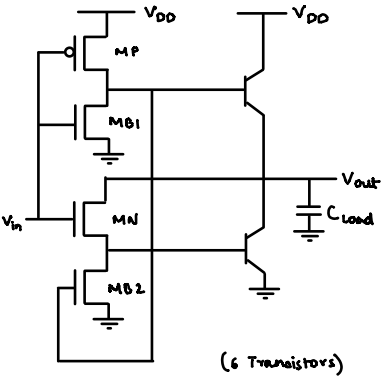


[1]

Totem Pole Configuration

MP - ON - Charges

MN - ON - Capacitor Discharges



(6 Transistors)

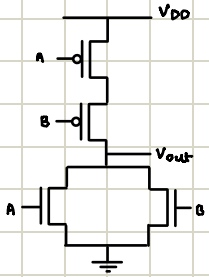
BiCMOS inverter with active base pull-down

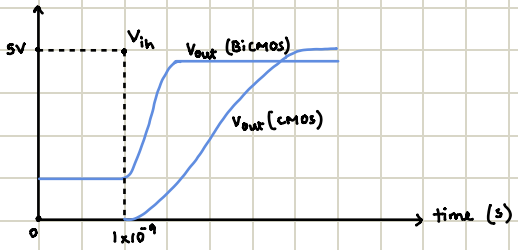
→ Easy to fabricate nMOS than resistor

→ The superiority of BiCMOS lies in the high current drive capability of BJT output, 0 static power dissipation, high input impedance provided by MOS

→ To reduce the turn off times of BJT during switching 2 minimum sized NMOS are added to provide the necessary base discharge path instead of resistor

→ 2 Input NOR gate CMOS

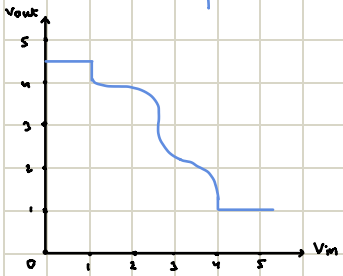


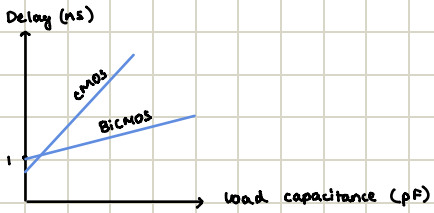


Simulated output voltage waveform of CMOS & BiCMOS for 5pF

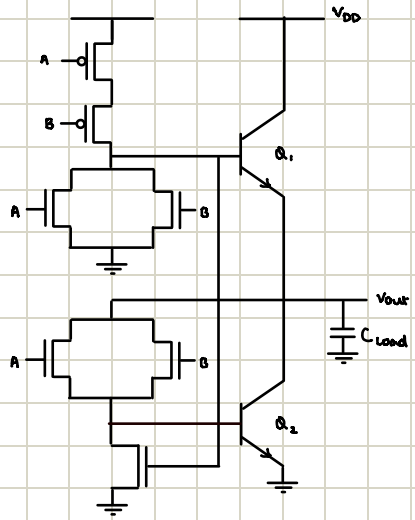
Both the circuits occupy approximately same amount of Si area

VTC curve of BiCMOS inverter

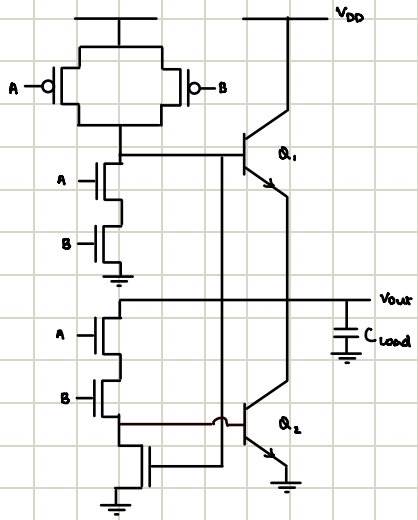




Q. 2 Input BiCMOS NOR gate



Q. 2 Input BiCMOS NAND gate

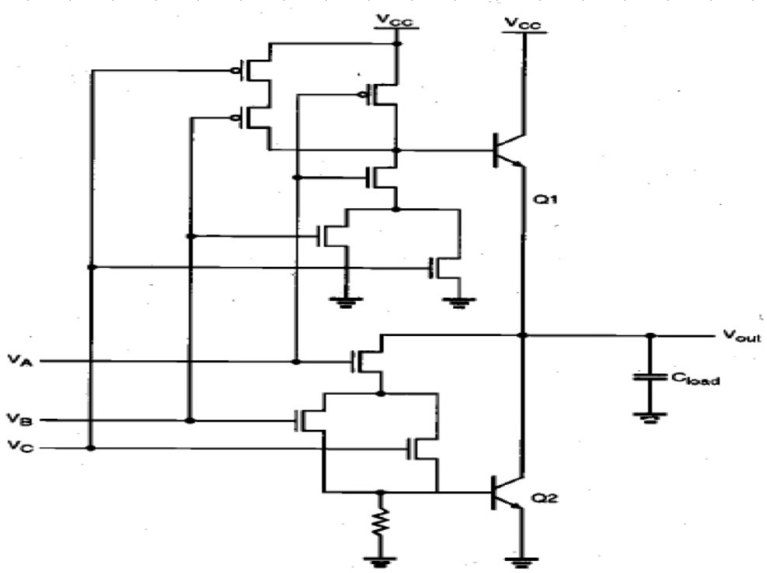


→ In BiCMOS circuit when pull-up PMOS devices are ON, it charges capacitor to V_{DD} (V_{out} is charged to V_{DD}), meanwhile base charge of Q_2 has to be removed by MB_2 NMOS

→ When pull-down NMOS are ON, it discharges load capacitance to 0 by turning Q_2 Meanwhile base charge of Q_1 will be discharged through NMOS transistors

Q. $Y = [(B+C).A]'$

A.



Unit - 2

Introduction

→ We discuss here about CMOS fabrication which requires NMOS & PMOS transistors be built on same chip substrate.

→ To accomodate nMOS & pMOS devices, special regions called wells are used for which semiconductor type is opposite to substrate type

(n-well created in p-type substrate
P-well created in n-type substrate)

Fabrication Process Flow

→ Fabrication is a process of creating an IC by sequence of photo-lithographic and chemical processing steps

Patterning

→ IC may be viewed as a set of patterned layers of doped silicon, polysilicon, metal & insulating SiO_2

→ A layer must be patterned before next layer of material is applied on the chip and the process of transferring a pattern to a layer on the chip is called lithography

→ Each layer has its own requirements and lithographic process is repeated for every step using different mask

→ The entire oxide surface is covered with a light-sensitive, acid-resistant organic polymer, initially insoluble which is called photoresist
They become soluble upon exposure to UV light

→ To selectively expose the photoresist, we have to cover some of the areas on the surface with a mask during exposure

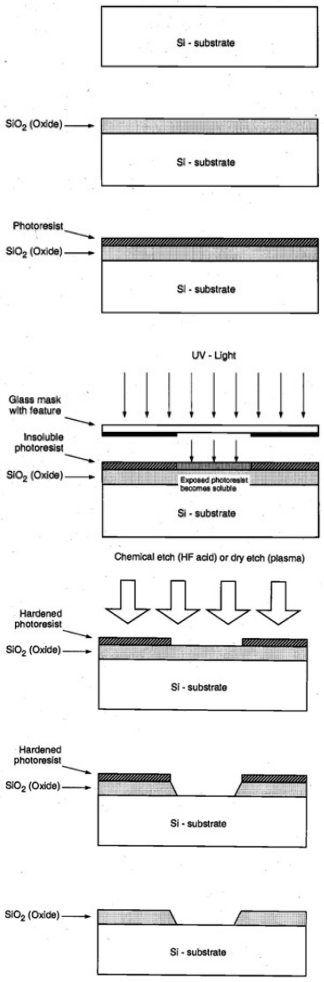
→ There are 2 types of photoresists:

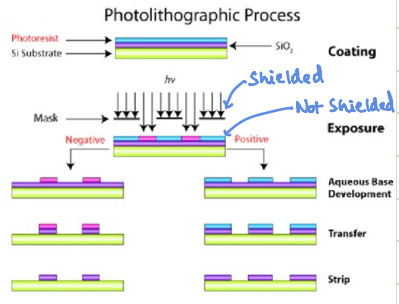
i) Positive Photoresist - Initially insoluble, soluble upon exposure to UV light

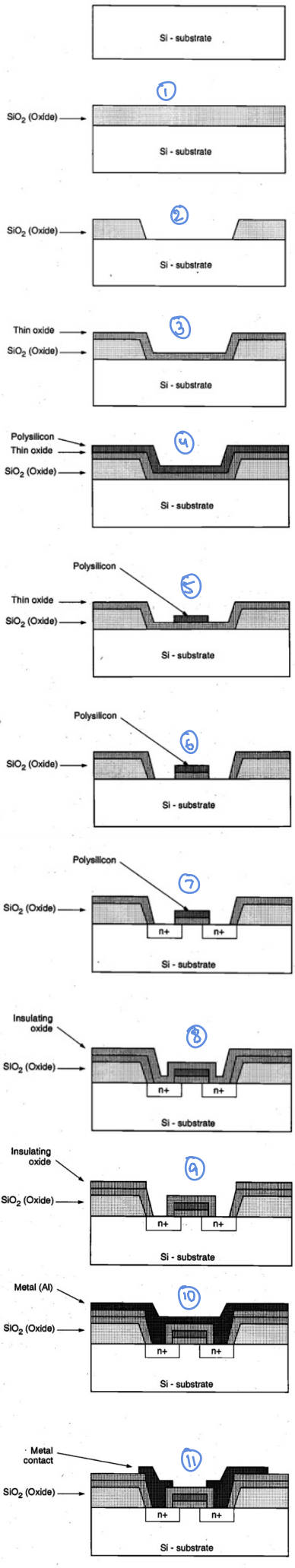
ii) Negative Photoresist - Initially soluble, insoluble upon exposure to UV light

(Negative photoresists are more sensitive to light but photolithographic resolution is not as high as that of positive photoresists)

Process steps required for patterning of Silicon Dioxide







Fabrication of nMOS Transistor

Step 1: Process starts with oxidation of Silicon substrate on which relatively thick SiO_2 layer (Field Oxide) is created on surface

Step 2: Field Oxide is selectively etched to expose silicon surface on which MOS transistor will be created

Step 3: Surface is covered with thin, high quality oxide layer, which eventually form gate oxide of MOS transistor

Step 4: On top of this, layer of polysilicon is deposited.
(Undoped polysilicon has high resistivity which can be reduced by doping with impurity atoms)

Step 5: Polysilicon layer is patterned & etched to form interconnects & MOS Transistor gates

Step 6: The thin gate which is exposed also gets etched, exposing bare silicon surface on which Source & Drain junctions are formed

Step 7: Entire Silicon surface is doped with high concentration impurities, by diffusion or ion implantation (Donor atoms produce n-type doping)

Step 8: Doping penetrates exposed areas, creating 2 n-type regions in p-type substrate which is again covered with an insulating layer of SiO_2

Step 9: Insulating oxide layer is patterned in order to provide contact windows for drain & source junctions

Step 10: Surface is covered with evaporated aluminium which will form interconnects

Step 11: Finally, metal layer is patterned & etched completing interconnection of MOS

Device Isolation Techniques

→ MOS Transistors consisting an IC must be electrically isolated from each other during fabrication to prevent unwanted conduction path between devices, avoiding creation of inversion layers outside channel region of transistors on a chip surface (called active area)

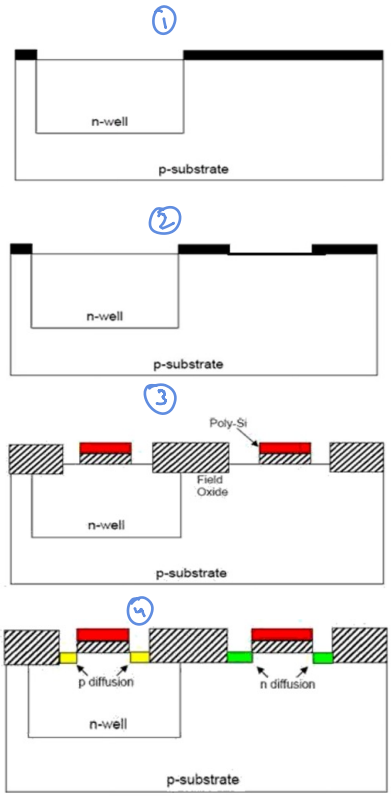
→ For NMOS Fabrication,
Silicon wafer → 75 to 150 mm in diameter & 0.4mm thick
Impurity concentrations → 10^{15} cm^{-3} (P-type impurities : Boron)
Resistivity → 2-25 Ωcm

Diffusion Implantation (Implantation of impurities to change resistivity)

→ In high temperature, dopant atoms move from high conc. to low conc. region
Disadvantage: sideways diffusion is difficult because small geometric

Ion Implantation

→ Dopant atoms implanted vertically into surface of silicon by high energy ion beam (eV), leading to less lateral diffusion



Fabrication of n-Well CMOS

Step 1: Moderately doped p-type substrate is prepared and lithographic mask is used to define N-well region which is formed by diffusion (or) ion implantation

Step 2: Active regions for NMOS & PMOS are defined using another lithographic mask which are used to determine where transistors will be built

Step 3: A thin SiO_2 layer is grown on wafer's surface which insulates gate & polysilicon layer is patterned using photolithography to form gate electrodes

Step 4: n-diffusions created in P-substrate for NMOS
p-diffusions created in N-well for PMOS

Step 5: Contact cuts are made to access diffusion regions & polysilicon layers

Step 6: Thin Aluminium layer is deposited & patterned to form interconnections

Step 7: Passivation layer is added for protection & Bonding pads for external connections

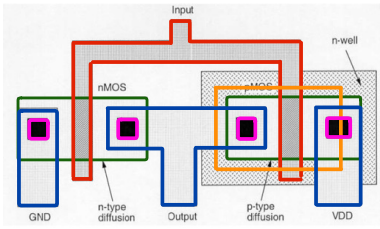
Layout Design Rules

→ Layout design is a schematic of the IC which describes the actual placement of devices / components on IC for fabrication

→ Layout design rules are the constraints to be followed to manufacture the physical mask layout of any circuit

→ The rules are :
- Information of dimensions of layers (metal, diffusion, polysilicon)
- How closely they can be placed (minimum width & minimum spacing constraints)

→ Objective :
IC should have high overall yield in smallest possible area



Active
Poly
Implant
Contact
Metal

Manufacturing Problems

→ Variation in V_{th} : Variation in ion-implantation & poly-layer

→ Diffusion, Poly, Metal : Variation in doping leading to variation in R & C

→ Shorts (2 lines placed too close) & Opens (Too small metal line may break)

→ Via/contact cuts problems : Consists of minimum-width & minimum spacing constraints & requirements b/w objects on same/different layers

→ Design rules are described in 2 ways :
i) Micron rules : Constraints are stated in terms of absolute dimensions in μm
ii) Lambda rules : Constraints are stated in terms of single parameter (λ)

Lambda Rules

→ λ is half of min. channel length ($L = 2\lambda$)

→ Industry layout design rules are generally in μ

→ All widths, spacing & distance are written if $m\lambda$ form (m : multiplier)

→ Change in λ can produce new mask

Summary of CMOS Fabrication

Formation of P-substrate & n-well → nMOS & pMOS active areas → Polysilicon Gate Formation → Source/Drain Diffusions → Contact Formation → Metal Layer Deposition → Passivation & Bonding Pads

Summary

AA ⇒ 1, 2 ⇒ 3λ

P.S ⇒ 3-7 ⇒ Mostly 2λ

Edge spacing

R6 outside 1λ

inside R7 3λ

Metal ⇒ 8, 9 ⇒ 3λ

P.C ⇒ 10 - 14 ⇒ Mostly 2λ

edge spacing

Poly & metal R12 ↓ R13 1λ

active ↓ R14 3λ

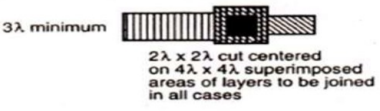
A.C ⇒ 15 - 20 ⇒ Mostly 2λ

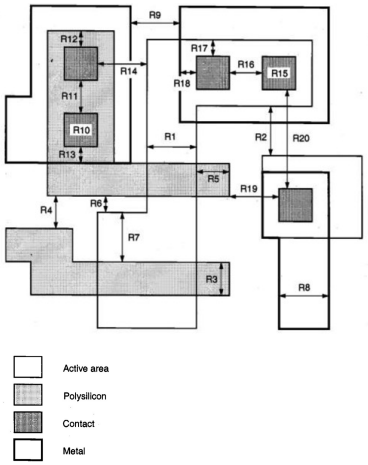
edge spacing

Active & metal R17 ↓ R18 1λ

Poly R19 ↓ 3λ

other active R20 ↘ 6λ





Active area rules

Rule No	Description	λ-Rule
R 1	Min. active area width	3λ
R 2	Min. active area spacing	3λ

Polysilicon rules

R 3	Min. poly width	2λ
R 4	Min. poly spacing	2λ
R 5	Min. gate extension of poly over active	2λ
R 6	Min. poly-active edge spacing (poly outside active area)	1λ
R 7	Min. poly-active edge spacing (poly inside active area)	3λ

Metal Rules

R 8	Min. metal width	3λ
R 9	Min. metal spacing	3λ

Metal 1 to polysilicon or to diffusion

→ 2 x 2 λ contact cut indicates an area in which oxide is to be removed down to the underlying polysilicon or diffusion surface

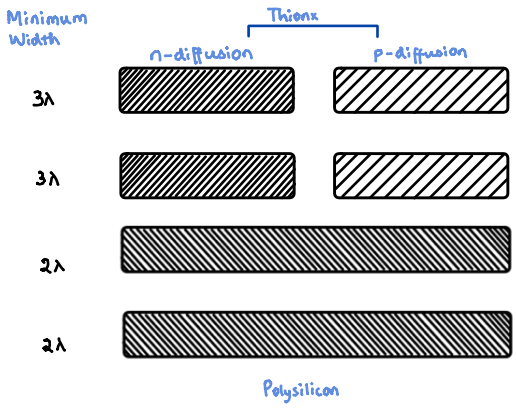
→ Deposition of metal layer takes place the metal is deposited through contact cut areas onto underlying area so that contact is made between the layers

Polycontact Rules

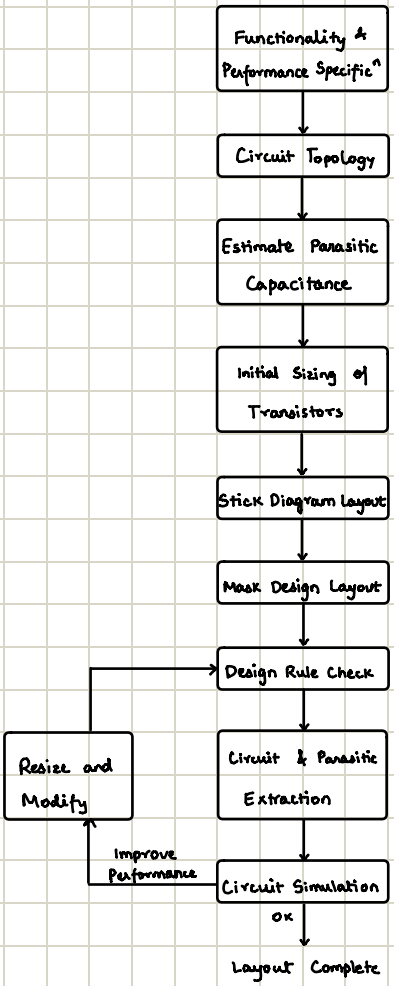
Rule No	Description	λ-Rule
R 10	Poly contact size	2λ
R 11	Minimum poly contact spacing	2λ
R 12	Minimum poly contact to poly edge spacing	1λ
R 13	Minimum poly contact to metal edge spacing	1λ
R 14	Minimum poly contact to active edge spacing	3λ

Active contact Rules

Rule No	Description	
R 15	Active contact size	2λ
R 16	Min. active contact spacing (on same active region)	2λ
R 17	Min. active contact to active edge spacing	1λ
R 18	Min. active contact to metal edge spacing	1λ
R 19	Min. active contact to poly edge spacing	3λ
R 20	Min. active contact spacing (on diff. active region)	6λ



Design Flow for production of a physical mask layout



FEATURE	FEATURE (STICK)
n-type enhancement mode transistor (as in Color plate 1(a))	Demarcation line, L: W
p-type enhancement mode transistor	L: W, S, G, D, Demarcation line

Rules

→ N/P diffusions should not cross the demarcation lines

→ N & P diffusions are interconnected through metal

→ Polysilicon & metal can cross the demarcation lines

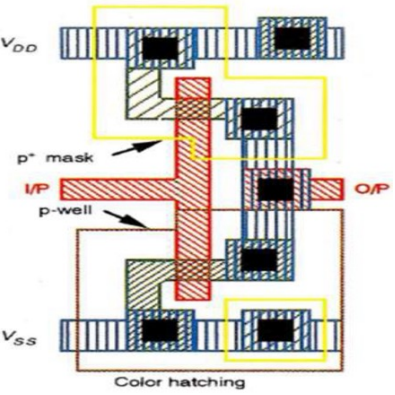
→ Crosses on V_{DD} / Grnd power supply rails indicate substrate / p-well connections

→ Diffusion paths are run parallel to V_{DD} & Ground

Additional Rules

R25	Minimum Active Overlap over contact	1.5λ
R26	Minimum Metal1 Overlap over contact	1λ
R27	Minimum Poly Overlap over contact	1.5λ
R28	Minimum spacing to gate	1λ

CMOS Inverter Layout



(·) operation
PMOS – Parallel
NMOS – Series

(+) Operation
PMOS – Series
NMOS – Parallel

Via Rules (via connects 2 metals)

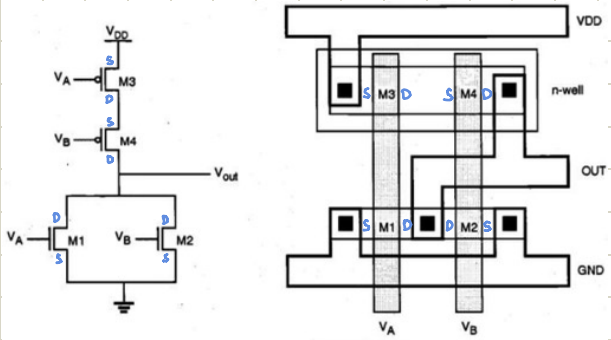
R29	Exact Size	2λ
R30	Minimum Spacing	3λ
R31	Minimum Overlap by Metal 1	1λ
R32	Minimum Spacing to contact	2λ
R33	Minimum Spacing to poly or active edge	2λ

Generic Combination Logic Circuit

→ OR operations performed by parallel-connected drivers
AND operations performed by series-connected drivers
→ Inversion is provided by nature of MOS Circuit
→ Each input variable is assigned to only 1 driver

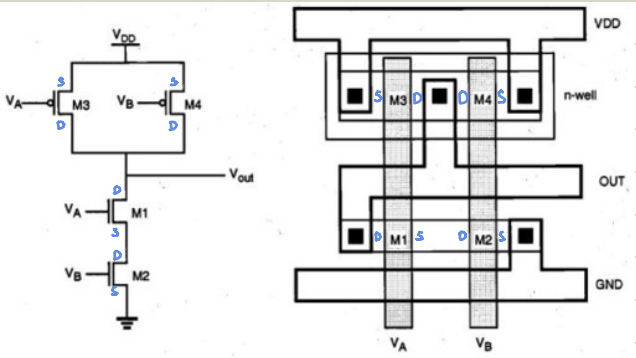
Complex Logic Circuits

→ NOR2



Layout Diagram Stick Diagram

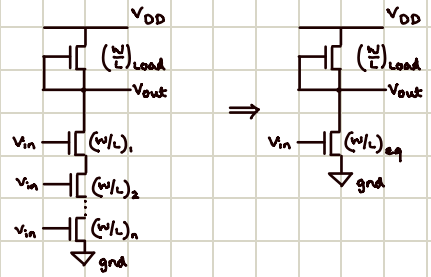
→ NAND2



Layout Diagram Stick Diagram

Generalized Structure & inverter equivalent

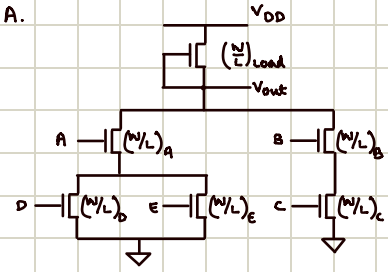
→ If we take a NAND Structure



$$I_D = \frac{\mu_n C_{ox}}{2}\left(\frac{1}{\sum_{K(on)}\frac{1}{\left(\frac{W}{L}\right)_K}}\right)\cdot\begin{cases}\left[2(V_{in}-V_{T_0})V_{out}-V_{out}^2\right], & \text{linear}\\ (V_{in}-V_{T_0})^2, & \text{saturation}\end{cases}$$

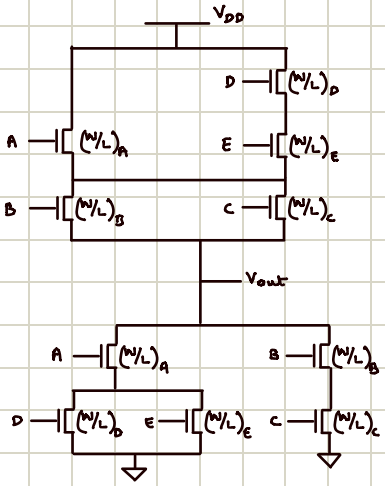
Q. $Z = \overline{A(D+E)+BC}$

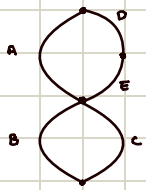
A.



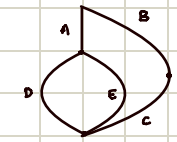
$$\left(\frac{W}{L}\right)_{eq} = \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_A}+\frac{1}{\left(\frac{W}{L}\right)_D+\left(\frac{W}{L}\right)_E}} + \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B}+\frac{1}{\left(\frac{W}{L}\right)_C}}$$

Euler Path for Graph

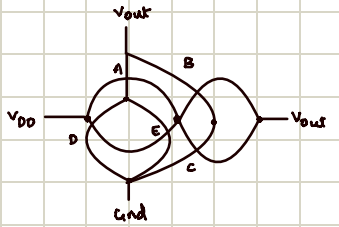




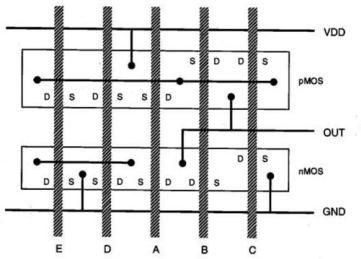
pMOS Network Graph



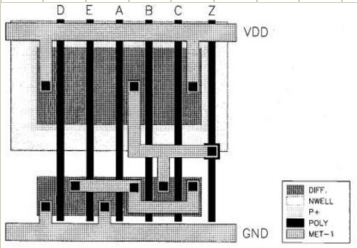
nMOS Network Graph



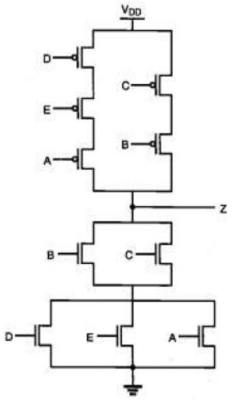
Optimized Stick Diagram



Q. Simplified layout of CMOS complex logic circuit is given. Draw corresponding circuit diagram & find equivalent CMOS inverter circuit for simultaneous switching of all inputs, assuming $\left(\frac{W}{L}\right)_p = 15$ & $\left(\frac{W}{L}\right)_n = 10$



A.



$$Z = \overline{(D+E+A)(B+C)}$$

$$\left(\frac{W}{L}\right)_{p,eq} = \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_D} + \frac{1}{\left(\frac{W}{L}\right)_E} + \frac{1}{\left(\frac{W}{L}\right)_A}} + \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B} + \frac{1}{\left(\frac{W}{L}\right)_C}}$$

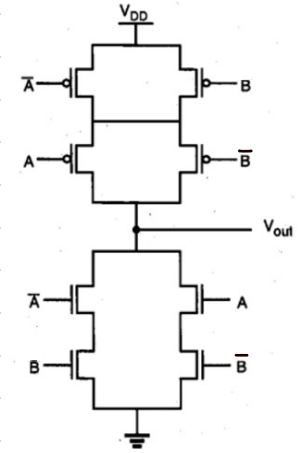
$$= \frac{1}{\frac{1}{15}+\frac{1}{15}+\frac{1}{15}} + \frac{1}{\frac{1}{15}+\frac{1}{15}} = 5 + 7.5 = 12.5$$

$$\left(\frac{W}{L}\right)_{n,eq} = \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B + \left(\frac{W}{L}\right)_C} + \frac{1}{\left(\frac{W}{L}\right)_D + \left(\frac{W}{L}\right)_E + \left(\frac{W}{L}\right)_A}}$$

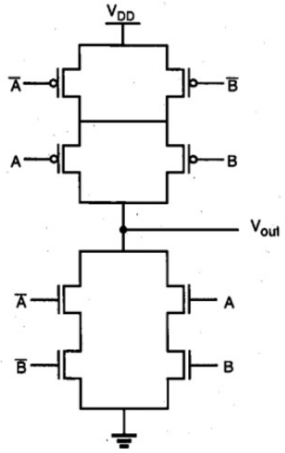
$$= \frac{1}{\frac{1}{10+10} + \frac{1}{10+10+10}} = \frac{600}{50} = 12$$

CMOS XOR & XNOR

A XOR B = $\bar{A}B + A\bar{B}$



A XNOR B = $\bar{A}\bar{B} + AB$

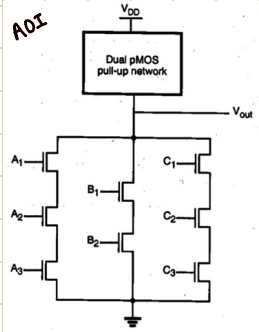


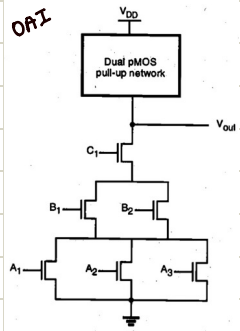
AOI and OAI Gates

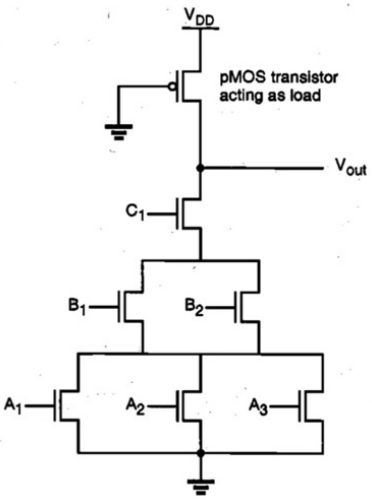
→ There are no strict limitations on topology of PUN & PDN but we have 2 types of gates :

i) AOI (AND-OR-INVERT) ⇒ Sum of products

ii) OAI (OR-AND-INVERT) ⇒ Product of sums







Pseudo NMOS Gates

→ A possible alternative to reduce number of transistors is to use single PMOS. with gate connected to ground as load device

→ This provides a simple pull-up arrangement, complex gate is implemented with fewer transistors

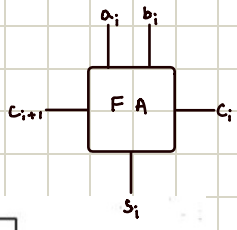
→ Disadvantage : Non-zero static power dissipation because of the always-ON PMOS load, which affects V_{OL} & noise margins based on transconductance ratio

CMOS Full-Adder

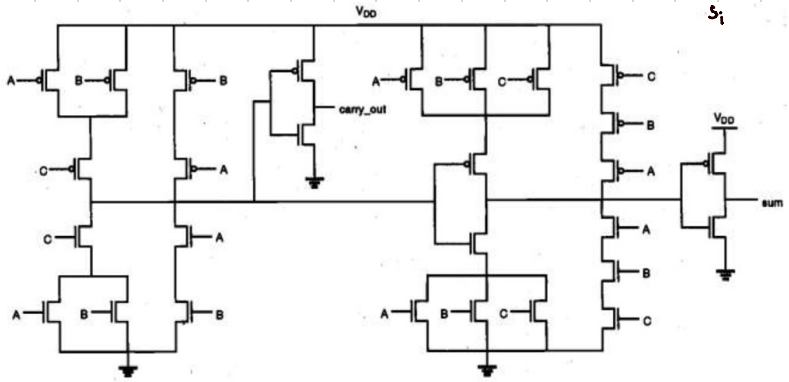
→ sum_out = $A \oplus B \oplus C$

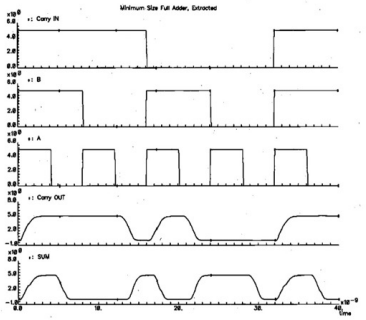
= $ABC + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C}$

carry_out = $AB + BC + AC$

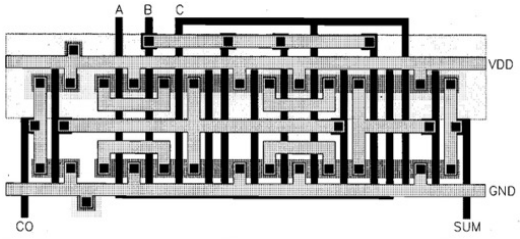


a_i	b_i	c_i	s_i	c_{i+1}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1





Simulated input & output waveforms of CMOS Full adder



Mask Layout of CMOS FA using min. size transistors

Year	Technology node	Shrink factor
1997	350 nm	250 ÷ 350 = 0.7142
1999	250 nm	180 ÷ 250 = 0.72
2001	180 nm	130 ÷ 180 = 0.7222
2003	130 nm	90 ÷ 130 = 0.6923
2006	90 nm	65 ÷ 90 = 0.7222
2008	65 nm	45 ÷ 65 = 0.6923
2010	45 nm	32 ÷ 45 = 0.7111
2012	32 nm	22 ÷ 32 = 0.6875

Year	1985	1987	1989	1991	1993	1995	1997	1999
Feature size (μm)	2.5	1.7	1.2	1.0	0.8	0.5	0.35	0.25

MOSFET Scaling & Small - Geometry Effects

→ Requirement to pack as many MOSFETs as possible in limited space (High Density)

→ The reduction of size is termed as scaling, which causes change in operational characteristics

→ Physical limitations limits the possibilities achievable by scaling

→ Scaling improves density, speed, power

→ We have 2 basic types of size-reduction strategies:

i) Full Scaling / Constant-field scaling

ii) Constant-Voltage Scaling

→ We use a constant called Scaling factor 's' ($s > 1$) to describe scaling

Constant - Field Scaling

→ Preserves magnitude of internal electric fields

i) Oxide Capacitance

$$C'_{ox} = \frac{\varepsilon_{ox}}{t'_{ox}} = S \cdot \frac{\varepsilon_{ox}}{t_{ox}} = S C_{ox}$$

ii) Drain Current

$$I'_D(\text{linear}) = \frac{1}{2}\mu_n C'_{ox}\frac{W'}{L'}\left(2(V'_{GS}-V'_T)V'_{DS} - V'^2_{DS}\right)$$

$$= \frac{1}{2}\mu_n S C_{ox}\frac{W}{L}\cdot\frac{S}{S}\left(2(V_{GS}-V_T)V_{DS} - V_{DS}^2\right)\cdot\frac{1}{S^2} = \frac{I_D}{S}(\text{linear})$$

Similarly $I'_D(\text{saturation}) = \frac{I_D}{S}(\text{saturation})$

iii) Power Dissipation

$$P' = I'_D \cdot V'_{DS} = \frac{1}{S^2}\cdot I_D V_{DS} = \frac{P}{S^2}$$

iv) Power Density

$$P'_D = \frac{P'}{A'} = \frac{P/S^2}{A/S^2} = \frac{P}{A} = P_D \quad (\text{Unchanged})$$

Constant - Voltage Scaling

→ Usually preferred over full scaling

	Quantity	Constant-field Scaling	Constant-Voltage Scaling (Preferred)
1)	Dimensions (L, W, t_{ox}, x_j)	Reduced by S	Reduced by S
2)	Voltages (V_{DD}, V_T)	Reduced by S	Unaffected
3)	Doping Densities (N_A, N_D)	Increased by S	Increased by S^2
4)	Drain Current (I_D)	Reduced by S	Increased by S
5)	Power Dissipation (P)	Reduced by S^2	Increased by S
6)	Power Density ($P_D = P/A$)	Unaffected	Increased by S^3
7)	Oxide Capacitance (C_{ox})	Increased by S	Increased by S

Advantages of Scaling

i) More capability

ii) Speed improvement

iii) Increase in current

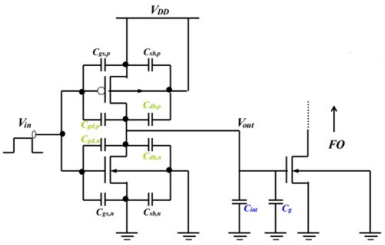
iv) Improved throughput

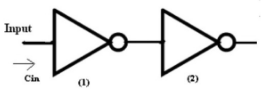
Disadvantages of Scaling

i) Short channel effects

ii) Complex process technology

iii) Parasitic effects dominate transistor effects





Cascaded CMOS Inverter Stage

→ Consider a cascode connection of 2 CMOS inverters

→ Capacitances

$C_{gd,p}$ & $C_{gd,n}$ are primarily due to gate overlap with diffusion

$C_{db,p}$ & $C_{db,n}$ are voltage dependant junction capacitances

C_g is due to thin-oxide capacitance over gate area

C_{int} is parasitic capacitance contribution of metal or polysilicon connection

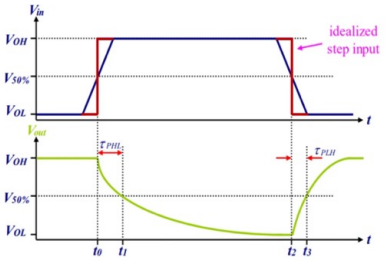
→ pulse waveform is applied to input of first stage inverter

For the circuit $C_{load} = C_{gd,n} + C_{gd,p} + C_{db,n} + C_{db,p} + C_{int} + C_g$

→ Ineffective parasitic capacitances

$C_{sb,n}$ & $C_{sb,p}$ because source-substrate voltage = 0

$C_{gs,n}$ & $C_{gs,p}$ because they are connected b/w input node & ground



Delay Definitions

i) Propagation Delay

τ_{PHL} : Time delay b/w $V_{50\%}$ transition of rising input to $V_{50\%}$ of falling output

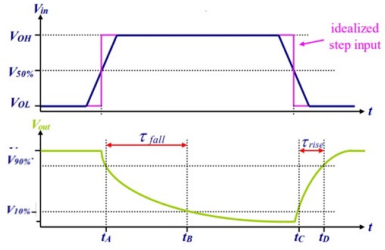
τ_{PLH} : Time delay b/w $V_{50\%}$ transition of falling input to $V_{50\%}$ of rising output

$$\tau_{PHL} = t_1 - t_0$$

$$\tau_{PLH} = t_3 - t_2$$

Avg. Propagation Delay: Avg. time required by input signal to propagate through inverter

$$\tau = \frac{\tau_{PHL} + \tau_{PLH}}{2}$$



ii) Rise time and Fall time

τ_r : Time required by output voltage to rise from 10% to 90%

τ_f : Time required by output voltage to fall from 90% to 10%

$$V_{10\%} = V_{OL} + 0.1(V_{OH} - V_{OL})$$

$$V_{90\%} = V_{OL} + 0.9(V_{OH} - V_{OL})$$

$$\tau_r = t_D - t_C$$

$$\tau_f = t_B - t_A$$

Delay Time Calculations

→ There are 2 methods

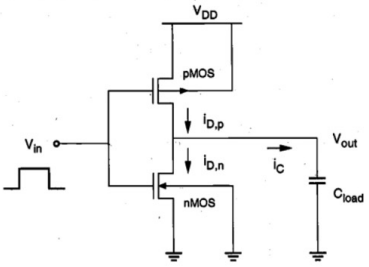
i) Average Current Method

ii) Differential Equation Method

→ Average Current Method

→ It is calculated during high-to-low transition of output by taking current values at beginning

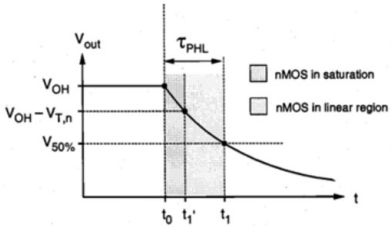
→ $i_C = C\frac{dv}{dt}$



$$\tau_{PHL} = \frac{C_{load}\cdot\Delta V_{HL}}{I_{avg,HL}} = \frac{C_{load}\,(V_{OH}-V_{50\%})}{I_{avg,HL}} \Rightarrow I_{avg,HL} = \frac{1}{2}\left[i_C(v_{in}=V_{OH}, v_{out}=V_{OH}) + i_C(v_{in}=V_{OH}, v_{out}=V_{50\%})\right]$$

$$\tau_{PLH} = \frac{C_{load}\cdot\Delta V_{LH}}{I_{avg,LH}} = \frac{C_{load}\,(V_{50\%}-V_{OL})}{I_{avg,LH}} \Rightarrow I_{avg,LH} = \frac{1}{2}\left[i_C(v_{in}=V_{OL}, v_{out}=V_{50\%}) + i_C(v_{in}=V_{OL}, v_{out}=V_{OL})\right]$$

→ Differential Equation Method



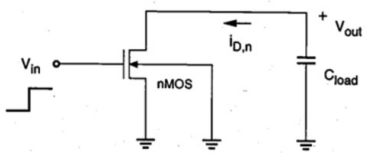
→ KCL at Output node, $C_{load}\frac{dV_{out}}{dt} = i_C = i_{D,p} - i_{D,n}$

$= -i_{D,n}$ (PMOS OFF, $i_{Dp} = 0$)

→ Consider NMOS in saturation from t_0 to t_1'

$V_{OH} - V_{T,n} < V_{out} \le V_{OH}$

$$i_{D,n} = \frac{K_n}{2}(V_{in} - V_{T,n})^2 = \frac{K_n}{2}(V_{OH} - V_{T,n})^2 \longrightarrow (1)$$



$$-i_{Dn} = C_{load}\frac{dV_{out}}{dt} \longrightarrow (2)$$

Solving (1) & (2),

$$\frac{K_n}{2}(V_{OH} - V_{T,n})^2 = -C_{load}\frac{dV_{out}}{dt}$$

$$\int_{t=t_0}^{t_1'} dt = -C_{load}\int_{V_{out}=V_{OH}}^{V_{OH}-V_{T,n}} \frac{dV_{out}}{i_{D,n}}$$

$$t_1' - t_0 = \frac{-2C_{load}}{K_n(V_{OH}-V_{T,n})^2}\left(V_{OH} - V_{T_n} - V_{OH}\right)$$

$$t_1' - t_0 = \frac{2V_{T,n}C_{load}}{K_n(V_{OH}-V_{T,n})^2}$$

$$\int \frac{dx}{2ax - x^2} = \frac{1}{2a} \ln\left(\frac{x}{2a - x}\right)$$

$a = V_{OH} - V_{T,n}$

$x = V_{out}$

$\ln(1) = 0$

$$V_{50\%} = \frac{V_{OL} + V_{OH}}{2}$$

→ Consider NMOS in saturation from t_1' to t_1

In a similar way, $$\int_{t=t_1'}^{t_1} dt = -2C_{load} \int_{V_{out}=V_{OH}-V_{T,n}}^{V_{50\%}} \left(\frac{1}{K_n\left(2(V_{OH}-V_{T,n})V_{out} - V_{out}^2\right)}\right) dV_{out}$$

$$t_1 - t_1' = \frac{-2C_{load}}{K_n} \int_{V_{OH}-V_{T,n}}^{V_{50\%}} \frac{dV_{out}}{2\left((V_{OH}-V_{T,n})V_{out} - V_{out}^2\right)}$$

$$= -\frac{C_{load}}{K_n} \cdot \frac{1}{(V_{OH}-V_{T,n})} \left[\ln\left(\frac{V_{out}}{2(V_{OH}-V_{T,n}) - V_{out}}\right)\right]_{V_{OH}-V_{T,n}}^{V_{50\%}}$$

$$= -\frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left[\ln\left(\frac{V_{50\%}}{2(V_{OH}-V_{T,n}) - V_{50\%}}\right) - \ln\left(\frac{V_{OH}-V_{T,n}}{2(V_{OH}-V_{T,n}) - (V_{OH}-V_{T,n})}\right)\right]$$

$$= \frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left[\ln\left(\frac{2(V_{OH}-V_{T,n}) - V_{50\%}}{V_{50\%}}\right)\right]$$

$$= \frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left[\ln\left(\frac{2V_{OH} - 2V_{Tn} - \frac{V_{OH}}{2} - \frac{V_{OL}}{2}}{\frac{V_{OH}+V_{OL}}{2}}\right)\right]$$

$$= \frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left[\ln\left(\frac{4(V_{OH}-V_{Tn})}{V_{OH}+V_{OL}} - 1\right)\right]$$

→ Now, $$t_{PHL} = (t_1' - t_0) + (t_1 - t_1')$$

$$= \left[\frac{2C_{load}V_{T,n}}{K_n(V_{OH}-V_{T,n})^2}\right] + \frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left[\ln\left(\frac{4(V_{OH}-V_{T,n})}{V_{OH}+V_{T,n}} - 1\right)\right]$$

$$= \frac{C_{load}}{K_n(V_{OH}-V_{T,n})} \left(\frac{2V_{T,n}}{V_{OH}-V_{Tn}} + \ln\left(\frac{4(V_{OH}-V_{Tn})}{V_{OH}+V_{T,n}} - 1\right)\right)$$

→ We can consider the same way for t_{PLH},

$$t_{PLH} = C_{load}\left[\int_{V_{out}=V_{OL}}^{V_{DD}-|V_{T,load}|} \frac{dV_{out}}{I_{D,load}(sat)} + \int_{V_{out}=V_{DD}-|V_{T,load}|}^{V_{50\%}} \frac{dV_{out}}{I_{D,load}(linear)}\right]$$

$$= \frac{C_{load}}{K_p(V_{DD}-|V_{T,p}|)} \left[\frac{2|V_{T,p}|}{V_{DD}-|V_{T,p}|} + \ln\left(\frac{4(V_{DD}-|V_{T,p}|)}{V_{DD}} - 1\right)\right]$$

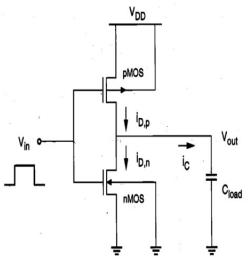
→ Actual time delay is calculated by considering step input as well

$$t_{PHL(actual)} = \sqrt{t_{PHL}^2 + \left(\frac{t_r}{2}\right)^2}$$

$$t_{PLH(actual)} = \sqrt{t_{PLH}^2 + \left(\frac{t_f}{2}\right)^2}$$

t_r : rise time

t_f : fall time



Q. Consider the CMOS inverter circuit shown in Fig. 6.2, with V_{DD} = 3.3 V. The *I-V* characteristics of the nMOS transistor are specified as follows: when V_{GS} = 3.3 V, the drain current reaches its saturation level I_{sat} = 2 mA for $V_{DS} \geq 2.5$ V. Assume that the input signal applied to the gate is a step pulse that switches instantaneously from 0 V to 3.3 V. Using the data above, calculate the delay time necessary for the output to fall from its initial value of 3.3 V to 1.65 V, assuming an output load capacitance of 300 fF.

A.

$t_{PHL} = t_{PHL}(t_1' \text{ to } t_0) + t_{PHL}(t_1 \text{ to } t_1)$

$I_{d,sat} = 2mA \quad \text{for} \quad V_{DS} \geq 2.5V$

$V_{GS} - V_{T_n} = 2.5$

$V_{T_n} = 3.3 - 2.5 = 0.8\ V$

$I_{d,sat} = \frac{K_n}{2}(V_{on} - V_{T_n})^2 \Rightarrow K_n = \frac{2 \times 2 \times 10^{-3}}{(3.3-0.8)^2} = 0.64\ mA/V^2$

$t_{PHL} = t_{PHL}(t_1' \text{ to } t_0) + t_{PHL}(t_1 \text{ to } t_1') = \left(\frac{2\,C_{load} \cdot V_{T_n}}{K_n(V_{OH}-V_{T_n})^2}\right) + \left(\frac{C_{load}}{K_n(V_{OH}-V_{T_n})}\right)\left(\ln\left(\frac{4(V_{OH}-V_{T_n})}{(V_{OH}+V_{OL})} - 1\right)\right)$

$= \frac{2 \times 300f \times 0.8}{0.64m\,(2.5)^2} + \frac{300f}{0.64m(2.5)} \ln\left(\frac{4(2.5)}{3.3} - 1\right)$

$= 120p + 132p = 252ps$

Q. For the CMOS inverter shown in Fig. 6.2 with a power supply voltage of V_{DD} = 5 V, determine the fall time τ_{fall}, which is defined as the time elapsed between the time point at which $V_{out} = V_{90\%} = 4.5$ V and the time point at which $V_{out} = V_{10\%} = 0.5$ V. Use both the average-current method and the differential equation method for calculating τ_{fall}. The output load capacitance is 1 pF. The nMOS transistor parameters are given as

$\mu_n C_{ox} = 20\mu A/V^2$

$\left(\frac{W}{L}\right)_n = 10$

$V_{T,n} = 1V$

A. Avg-current method,

$I_{avg} = \frac{1}{2}\left[I(V_{in}=5, V_{out}=4.5) + I(V_{in}=5V, V_{out}=0.5V)\right]$

$= \frac{1}{2}\left[\frac{1}{2}K_n(V_{in}-V_{T_n})^2 + \frac{1}{2}K_n\left(2(V_{in}-V_{T_n})V_{out} - V_{out}^2\right)\right]$

$= \frac{1}{2} \times \frac{1}{2} \times 20 \times 10^{-6} \times 10\left[(5-1)^2 + (2(5-1)0.5 - 0.5^2)\right] = 0.9875\ mA$

$\tau_{fall} = \frac{C\Delta V}{I_{avg}} = \frac{1 \times 10^{-12}(4.5-0.5)}{0.9875 \times 10^{-3}} = 4.05ns$

Differential equation method,

$C\frac{dV_{out}}{dt} = \frac{-1}{2}K_n(V_{in}-V_{T,n})^2 \qquad \left(K_n = \mu_n C_{ox}\left(\frac{W}{L}\right)_n\right)$

$\frac{dV_{out}}{dt} = \frac{-20 \times 10^{-6} \times 10\,(5-1)^2}{2 \times 1 \times 10^{-12}} = -1.6 \times 10^9\ V/s$

$t_{sat} = \frac{-1}{1.6 \times 10^9}\int_{4.5}^{4} dV_{out} = 0.3125\ ns$

$C\frac{dV_{out}}{dt} = \frac{-1}{2}K_n\left(2(V_{in}-V_{T_n})V_{out} - V_{out}^2\right) \Rightarrow t_{fall} - t_{sat} = -2C\int_4^{0.5} \frac{dV_{out}}{K_n\left(2(V_{in}-V_{T,n})V_{out} - V_{out}^2\right)}$

$t_{linear} = \frac{C}{K_n} \cdot \frac{1}{(V_{in}-V_{T_n})}\left(\frac{2(V_{OH}-V_{T,n}) - V_{50\%}}{V_{50\%}}\right)_4^{0.5} = \frac{10^{-12}}{20 \times 10^{-6} \times 10 \times 4}\left(\ln\left(\frac{8-0.5}{0.5}\right) - \ln\left(\frac{8-4}{4}\right)\right) = 3.385ns$

$t_{fall} = t_{linear} + t_{sat} = 3.385 + 0.3125 = 3.6975\ ns$

Symmetric CMOS Inverter

→ We know,

$$V_{Th} = \frac{V_{Ton} + \sqrt{\frac{1}{K_R}}\left(V_{DD} + V_{Top}\right)}{1 + \sqrt{\frac{1}{K_R}}}$$

$$\sqrt{\frac{1}{K_R}} = \frac{V_{Th} - V_{Ton}}{V_{DD} + V_{Top} - V_{Th}}$$

$$K_R = \left(\frac{K_n}{K_p}\right) = \left(\frac{V_{DD} + V_{Top} - V_{Th}}{V_{Th} - V_{Ton}}\right)^2$$

$$\left(\frac{K_n}{K_p}\right)_{ideal} = \left(\frac{0.5V_{DD} + V_{Top}}{0.5V_{DD} - V_{Ton}}\right)^2 \qquad \left(V_{Th,ideal} = 0.5V_{DD}\right)$$

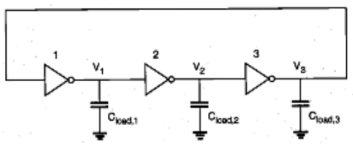
For symmetric CMOS, $\left(\frac{K_n}{K_p}\right) = 1$ $\qquad \left(|V_{Top}| = V_{Ton} = V_{To}\right)$

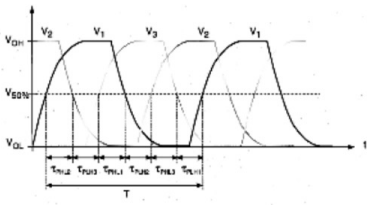
→ $V_{IL} = \frac{3V_{DD} + 2V_{Ton}}{8}$ $\qquad \left(V_{IL} = \frac{2V_{out} + V_{Top} - V_{DD} + K_R V_{Ton}}{1 + K_R}\right)$

$V_{IH} = \frac{5V_{DD} - 2V_{Ton}}{8}$

$V_{IL} + V_{IH} = V_{DD}$

Ring Oscillator





→ The cascade connection of 3 (can be any odd no.) identical CMOS inverters and output node of 3rd inverter is connected to input node of first forming a feedback loop

→ The circuit oscillates once any input or output voltage deviates from unstable operating point V_{Th}

→ 3 Stage oscillation period T

$$T = \tau_{PHL_1} + \tau_{PLH_1} + \tau_{PHL_2} + \tau_{PLH_2} + \tau_{PHL_3} + \tau_{PLH_3}$$

$$= 2\tau_p + 2\tau_p + 2\tau_p = 6\tau_p$$

$$f = \frac{1}{2n\tau_p}$$

→ Ring oscillator can be used as very simple pulse generator

Output waveform is utilized as simple master CLOCK signal generated on chip